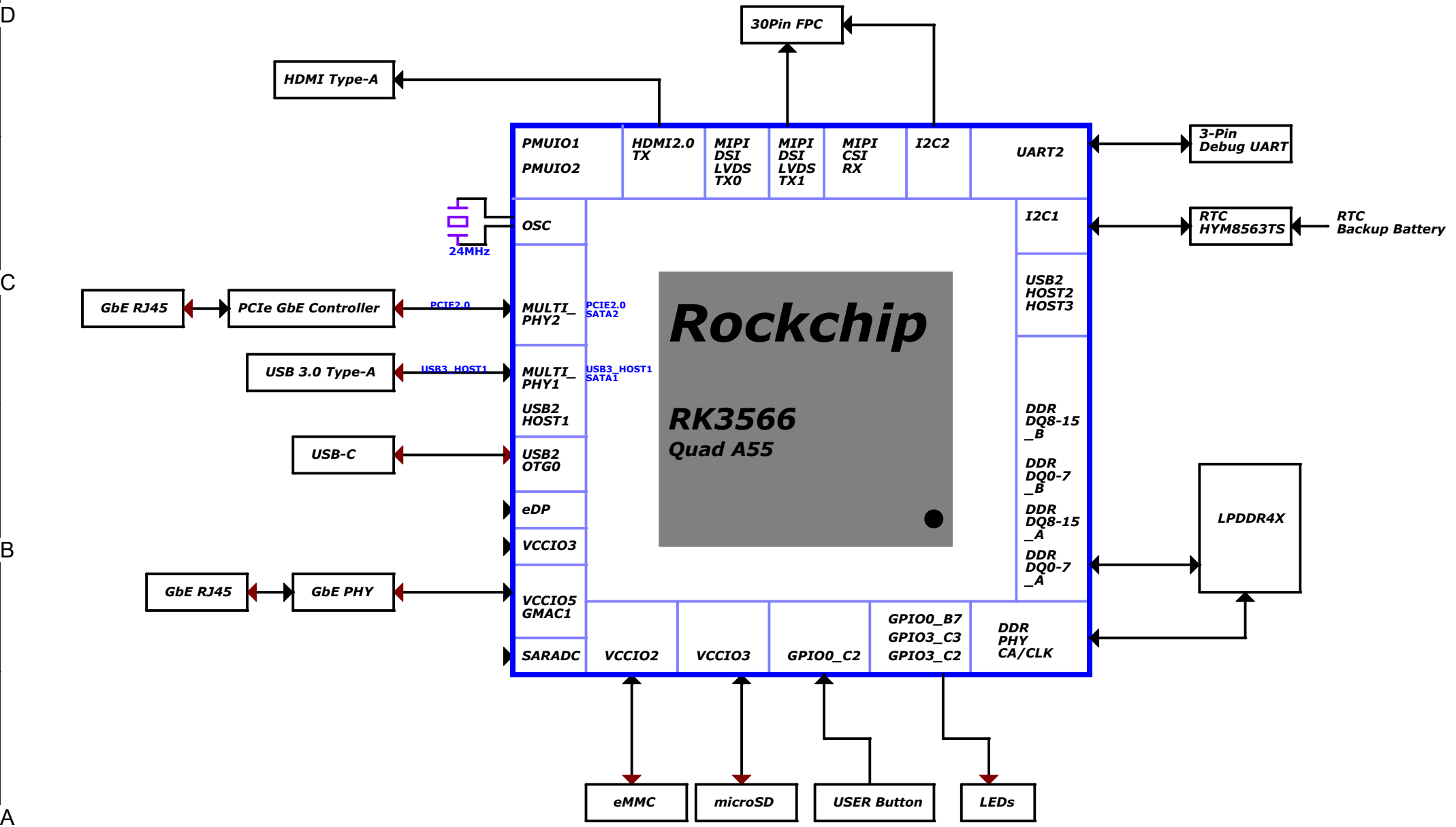
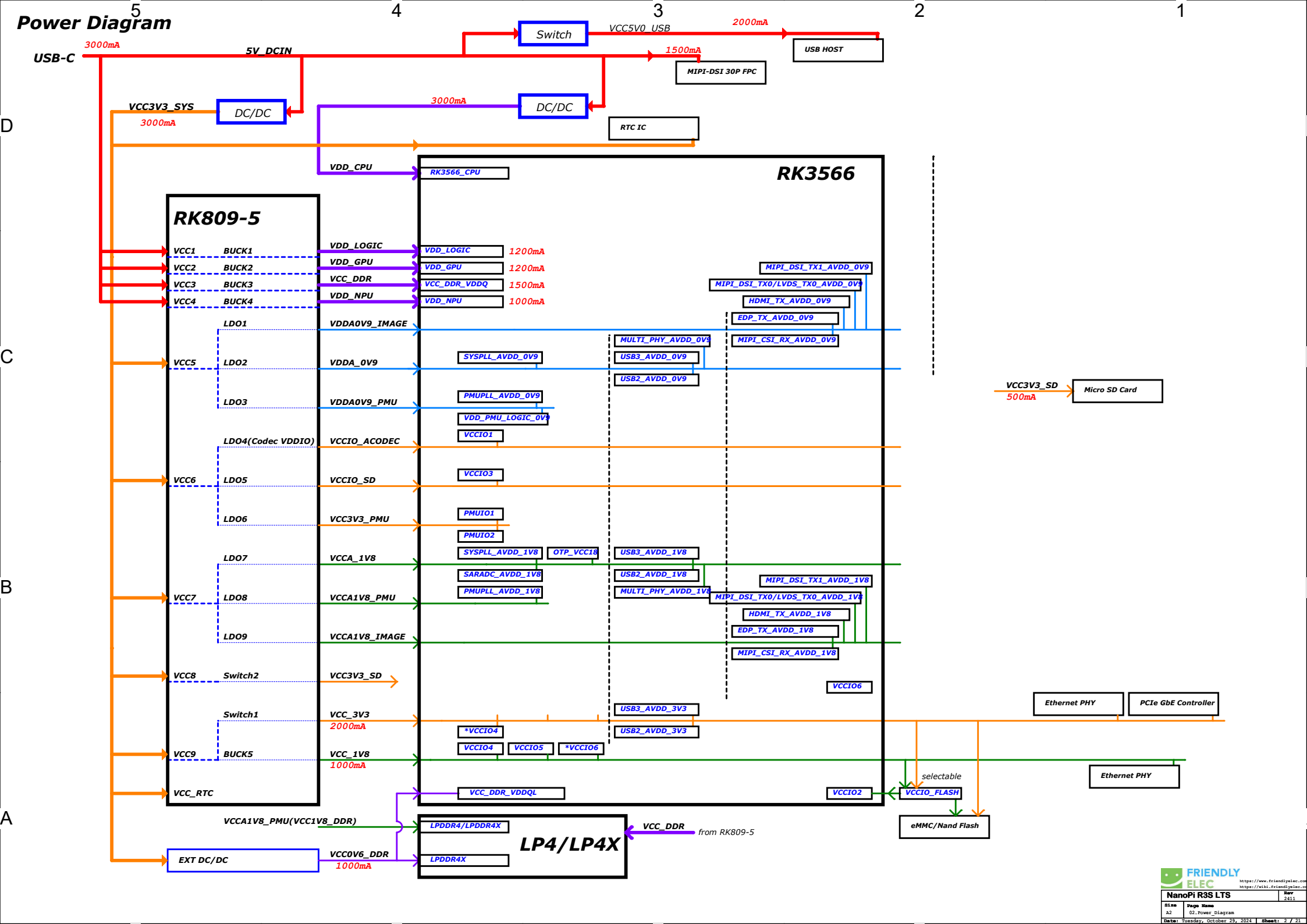
# NanoPi R3S LTS




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https://wiki.friendlyelec.com/

| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 01.Block_Diagram | |
| Date: Tuesday, October 29, 2024 | | Sheet: 1 / 21 |

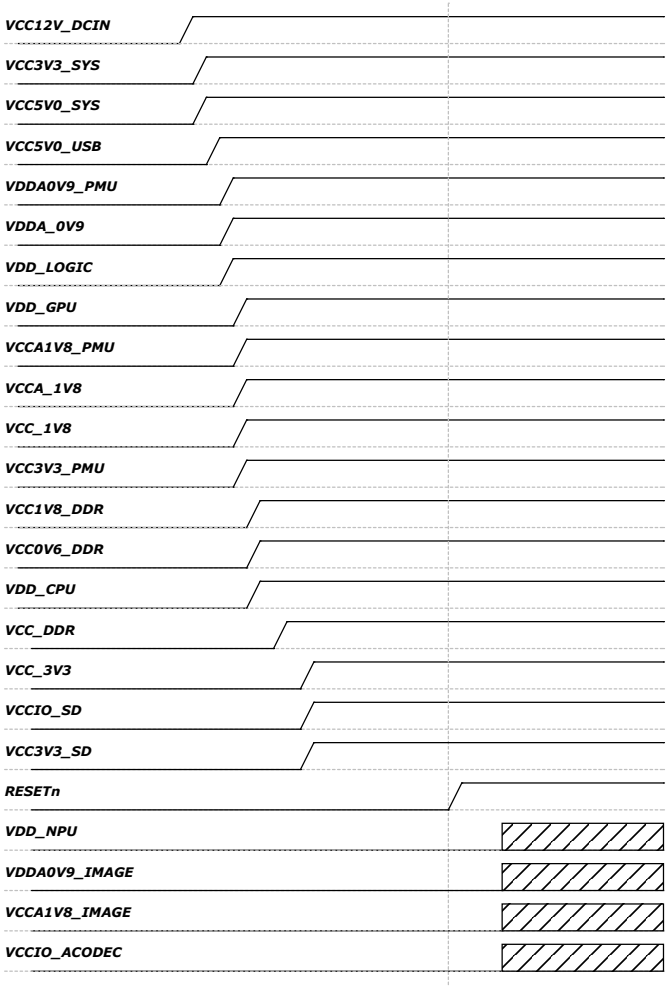
# Power Diagram




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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A2 | Page Name 02.Power_Diagram | |
| Date: Tuesday, October 29, 2024 | Sheet: 2 / 21 | |

# Power Sequence &Power Path assignment




| Power Source | PMIC Channel | Supply Limit | Power Supply Name | Time Slot | Default Voltage | Work Status | Sleep Status |
|---|---|---|---|---|---|---|---|
| VCC3V3_SYS | RK809_BUCK1 | 2.5A | VDD_LOGIC | Slot:1 | 0.9V | ON | OFF |
| VCC3V3_SYS | RK809_BUCK2 | 2.5A | VDD_GPU | Slot:2 | 0.9V | ON | OFF |
| VCC3V3_SYS | RK809_BUCK3 | 1.5A | VCC_DDR | Slot:3 | ADJ FB=0.8V | ON | ON |
| VCC3V3_SYS | RK809_BUCK4 | 1.5A | VDD_NPU | N/A | --/0.9V | OFF | OFF |
| VCC3V3_SYS | RK809_LDO1 | 0.4A | VDDA0V9_IMAGE | N/A | --/0.9V | OFF | OFF |
| | RK809_LDO2 | 0.4A | VDDA_0V9 | Slot:1 | 0.9V | ON | OFF |
| | RK809_LDO3 | 0.1A | VDDA0V9_PMU | Slot:1 | 0.9V | ON | ON |
| VCC3V3_SYS | RK809_LDO4 | 0.4A | VCCIO_ACODEC | N/A | --/3.3V | OFF | OFF |
| | RK809_LDO5 | 0.4A | VCCIO_SD | Slot:4 | 3.3V | ON | OFF |
| | RK809_LDO6 | 0.4A | VCC3V3_PMU | Slot:2 | 3.3V | ON | ON |
| VCC3V3_SYS | RK809_LDO7 | 0.4A | VCCA_1V8 | Slot:2 | 1.8V | ON | OFF |
| | RK809_LDO8 | 0.4A | VCCA1V8_PMU | Slot:2 | 1.8V | ON | ON |
| | RK809_LDO9 | 0.4A | VCCA1V8_IMAGE | N/A | --/1.8V | OFF | OFF |
| VCC3V3_SYS | RK809_SW2 100mohm | 2.1A | VCC3V3_SD | Slot:4 | 3.3V | ON | OFF |
| VCC3V3_SYS | RK809_SW1 90mohm | 2.1A | VCC_3V3 | Slot:4 | 3.3V | ON | OFF |
| | RK809_BUCK5 | 2.5A | VCC_1V8 | Slot:2 | 1.8V | ON | OFF |
| | RK809_RESETn | | | Slot:4+5 | | | |
| VCC12V_DCIN | EXT BUCK | 3.0A | VCC3V3_SYS | Slot:0 | 3.3V | ON | ON |
| VCC12V_DCIN | EXT BUCK | 3.0A | VCC5V0_SYS | Slot:0 | 5.0V | ON | ON |
| VCC5V0_SYS | EXT BUCK | 6.0A | VDD_CPU | Slot:2A | 1.025V | ON | OFF |
| | | | | | | | |
| | | | | | | | |

# IO Power Domain Map

## Refer to the actual design!

| IO Domain | Pin Num | Support IO Voltage | | Assignment IO Domain Voltage | | | Notes |
|---|---|---|---|---|---|---|---|
| | | 3.3V | 1.8V | Supply Power Net Name | Power Source | Voltage | |
| PMUIO1 | 1P16 | YES | NO | VCC3V3_PMU | VCC3V3_PMU | 3.3V | |
| PMUIO2 | 1N15 | YES | YES | VCC3V3_PMU | VCC3V3_PMU | 3.3V | 3.3V as default, dts config should follow the HW design |
| VCCIO1 | 1D13 | YES | YES | VCCIO_ACODEC | VCCIO_ACODEC | 3.3V | |
| VCCIO2 | 1C13 | YES | YES | VCCIO_FLASH | VCC_1V8 | 1.8V | FLASH_VOL_SEL = 1 --> VCCIO_FLASH = 1.8V |
| VCCIO3 | 1F17 | YES | YES | VCCIO_SD | VCCIO_SD | 3.3V | |
| VCCIO4 | 1E16 | YES | YES | VCCIO_WL | VCC_1V8 | 1.8V | |
| VCCIO5 | 1N5 1N6 | YES | YES | VCCIO5 | VCC_1V8 | 1.8V | VCC_1V8 as default, VCC_3V3 optional |
| VCCIO6 | 1L4 1L5 | YES | YES | VCCIO6 | VCC1V8_DVP | 1.8V | VCC1V8_DVP as default, VCC_1V8 optional |
| VCCIO7 | 1N8 | YES | YES | VCCIO7 | VCC_3V3 | 3.3V | |

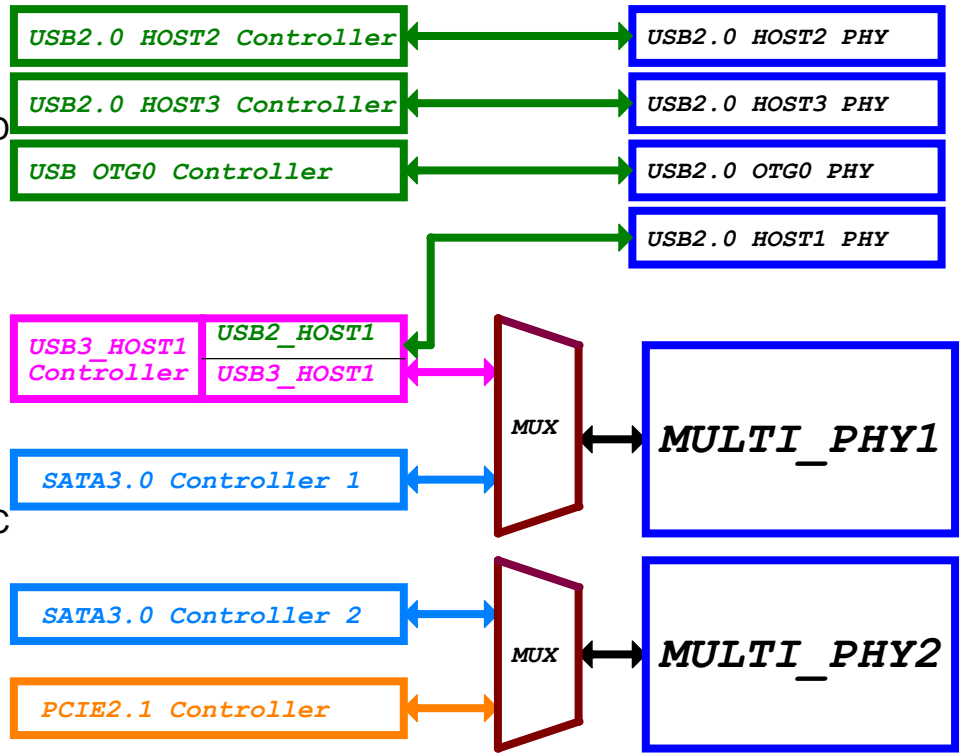
软件dts的电压配置严格与硬件设计保持一致！
Check the software configuration(dts) of voltage level, which must be keep the same as hardware design.

**!!! Attention 注意**

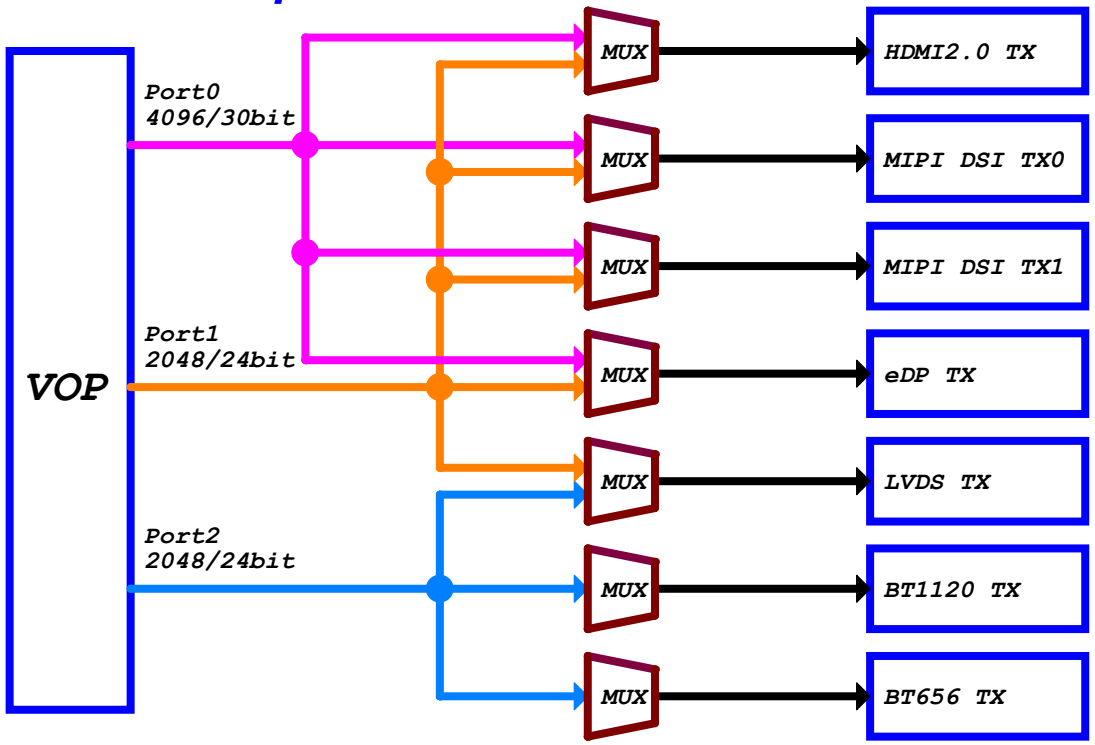
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| NanoPi R3S LTS | Rev 2411 |
|---|---|
| Size A3 | Page Name 03.Power_Sequence/IO_Domain_Map |
| Date: Tuesday, October 29, 2024 | Sheet: 3 / 21 |

## MULTI_PHY1/2 Path Map

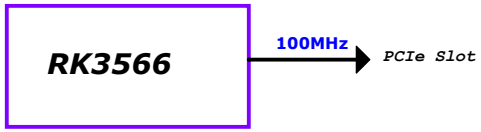



## VOP Path Map




RK3566 support max to 2 output display ports! !

## PCIe2.1 REFCLK-RC Mode




## PCIe2.1 PHY

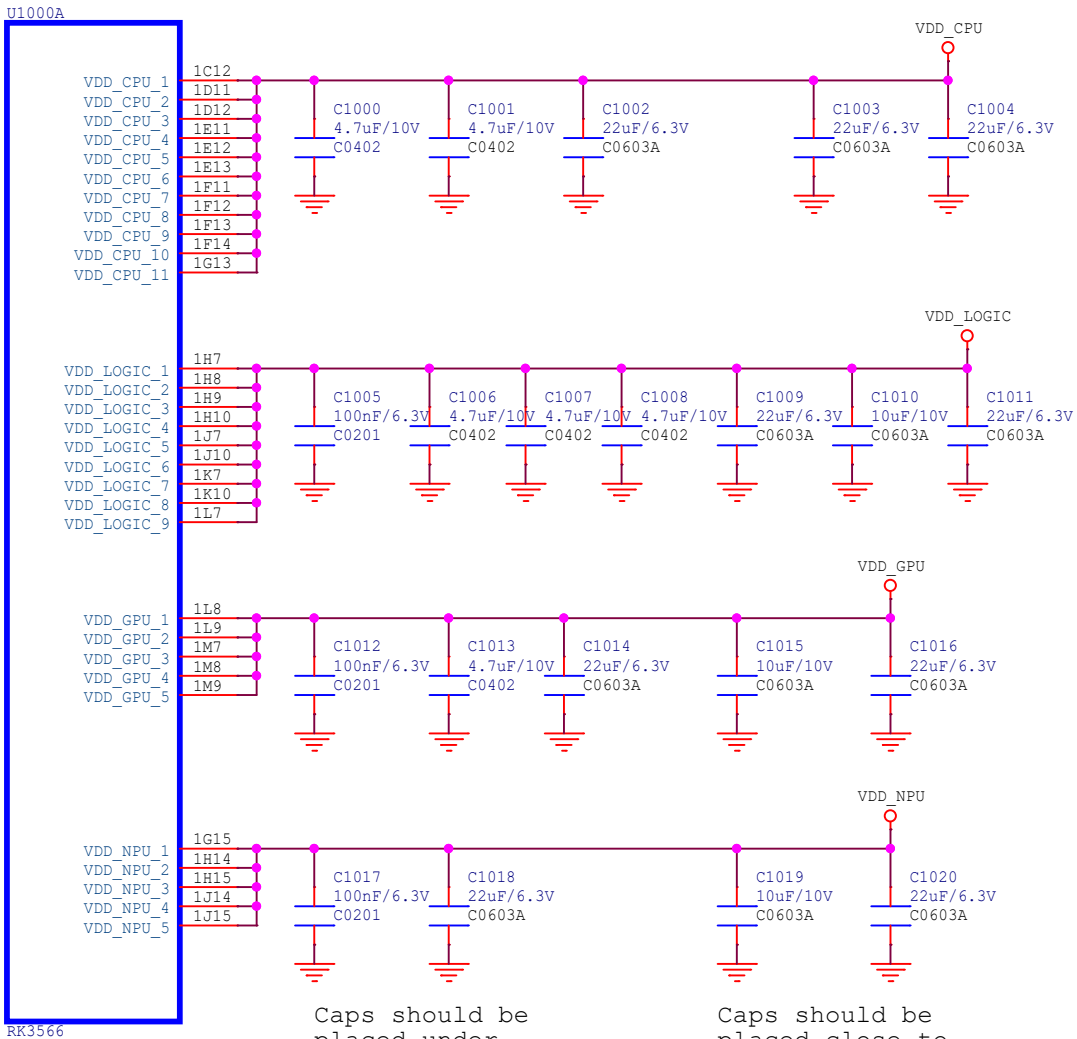
| | | | | | |
|---|---|---|---|---|---|
| MULTI_ PHY2 | PCIe2.1 x1Lane | PCIE20_REFCLK (RC:output) | PCIE20_TX<br>PCIE20_RX | PCIE20_CLKREQn<br>PCIE20_WAKEn<br>PCIE20_PERSTn<br>PCIE20_BUTTONRSTn | Only RC |

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| | |
|---|---|
| NanoPi R3S LTS | Rev 2411 |
| Size A3 | Page Name 04.MULTI_PHY/VOP_Fun_Map |
| Date: Tuesday, October 29, 2024 | Sheet: 4 / 21 |

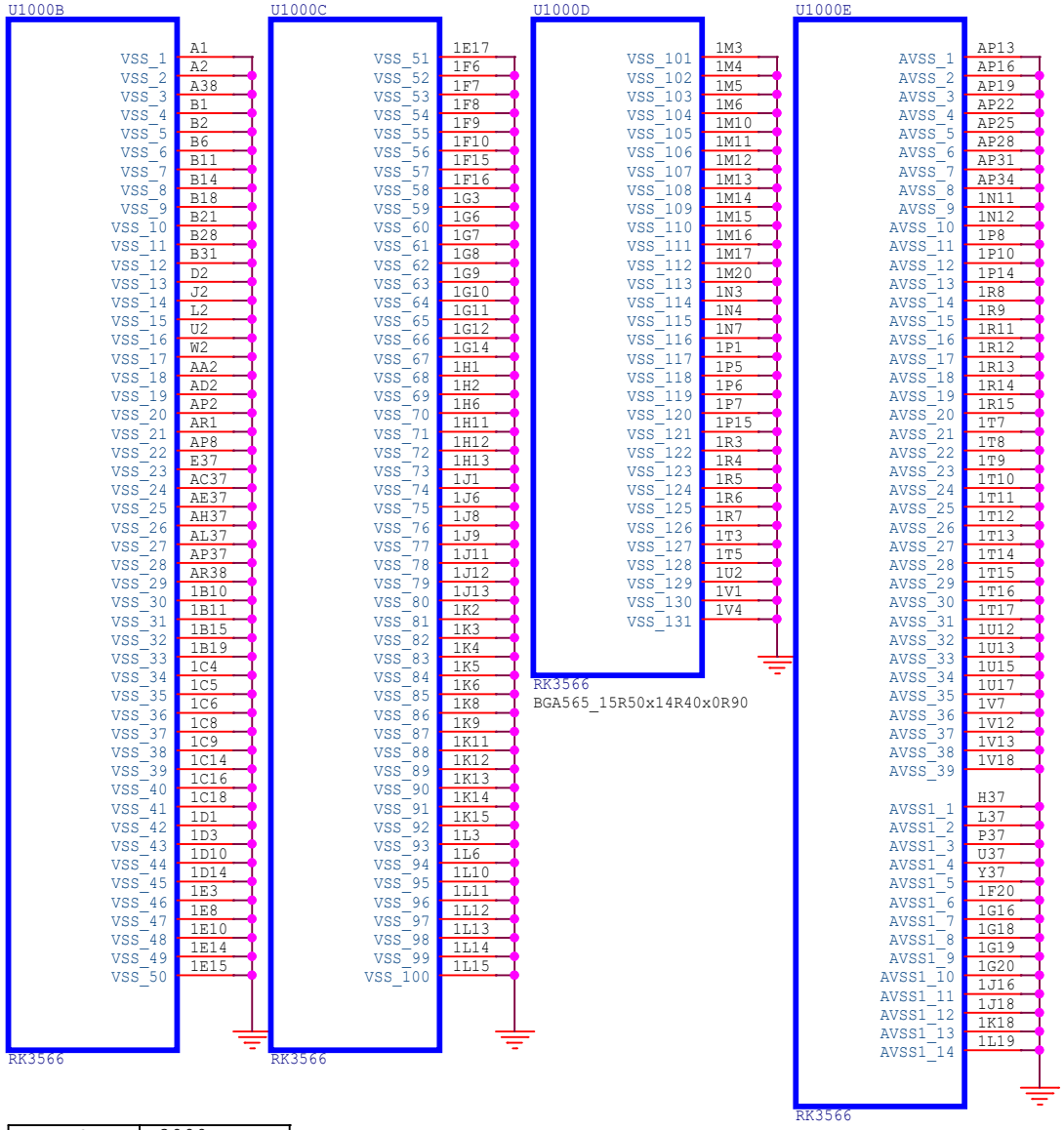
# RK3566_ABCDE

# (Power&GND)




Caps should be placed under the U1000 package

Caps should be placed close to the U1000 package



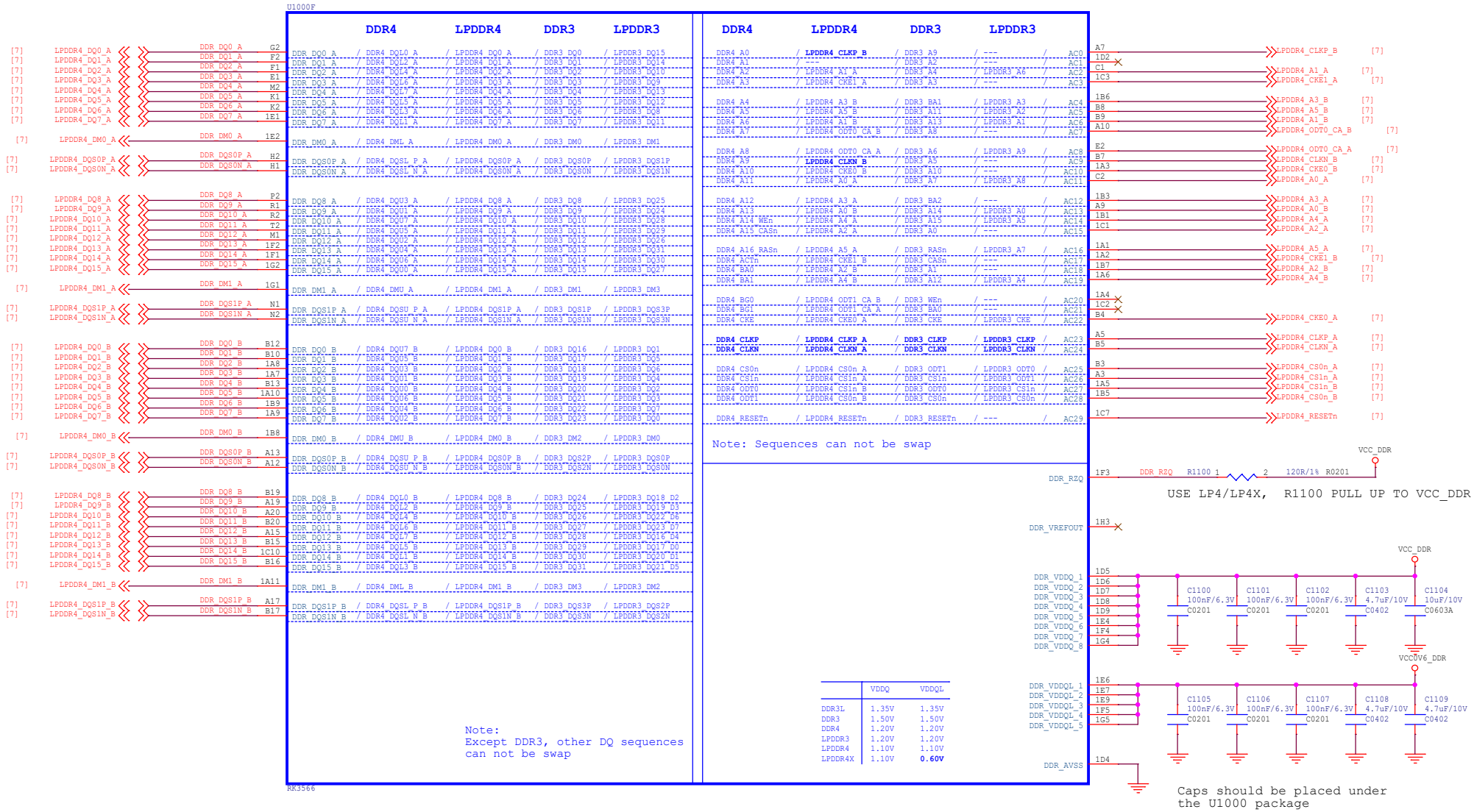

| | |
|---|---|
| VDD_CPU | 3000mA |
| VDD_GPU | 1200mA |
| VDD_NPU | 1000mA |
| VDD_LOG | 1200mA |

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| NanoPi R3S LTS | Rev 2411 |
|---|---|
| Size A4 | Page Name 05.RK3566_Power/GND |
| Date: Tuesday, October 29, 2024 | Sheet: 5 / 21 |

# RK3566_F(DDR PHY)




| | VDDQ | VDDQL |
|---|---|---|
| DDR3L | 1.35V | 1.35V |
| DDR3 | 1.50V | 1.50V |
| DDR4 | 1.20V | 1.20V |
| LPDDR3 | 1.20V | 1.20V |
| LPDDR4 | 1.10V | 1.10V |
| LPDDR4X | 1.10V | **0.60V** |

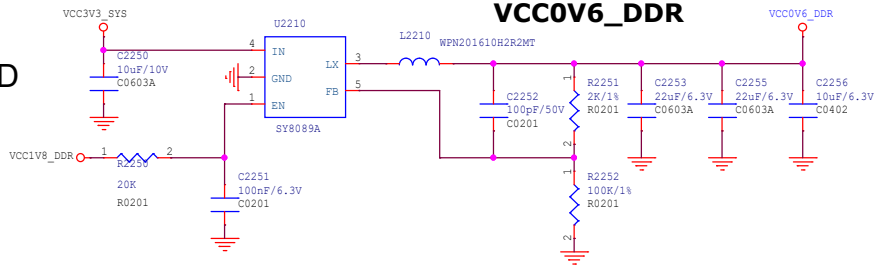
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| NanoPi R3S LTS | Rev 2411 |
|---|---|
| Size A3 | Page Name 06.RK3566_DDR PHY |
| Date: Tuesday, October 29, 2024 | Sheet: 6 / 21 |

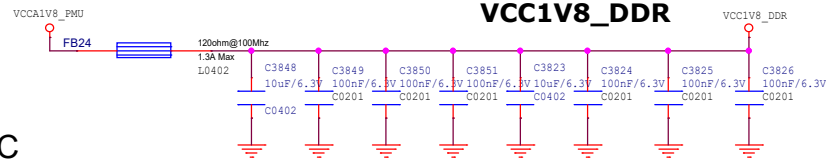
For LPDDR4: R2251 -> 84.5K

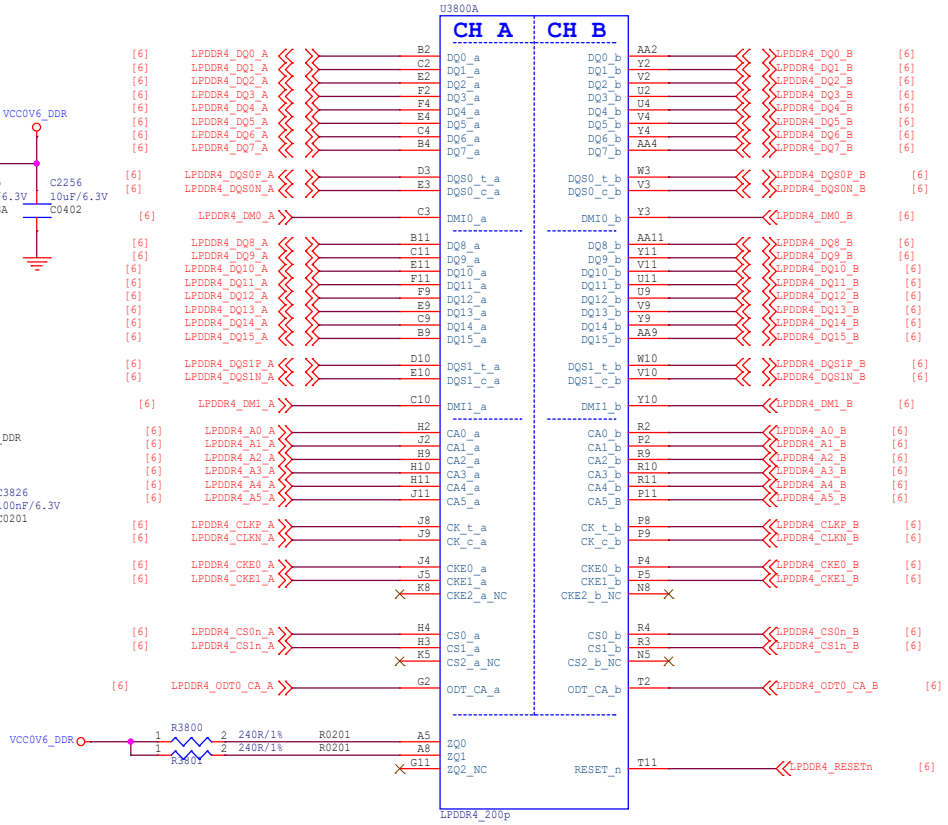
For LPDDR4X: R2251 -> 2K

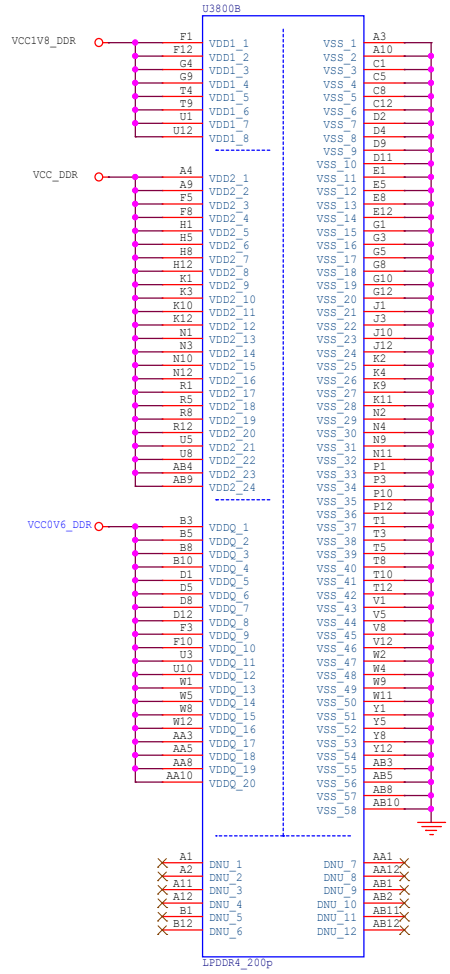
## VCC0V6_DDR

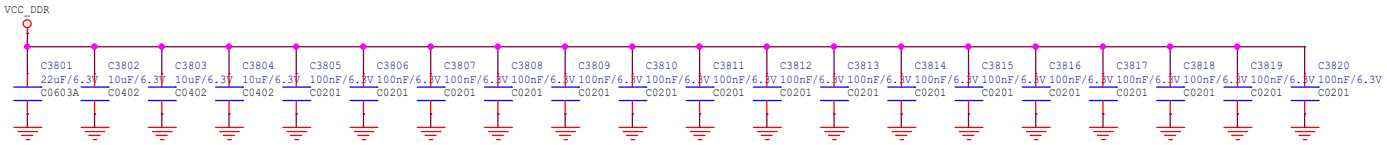


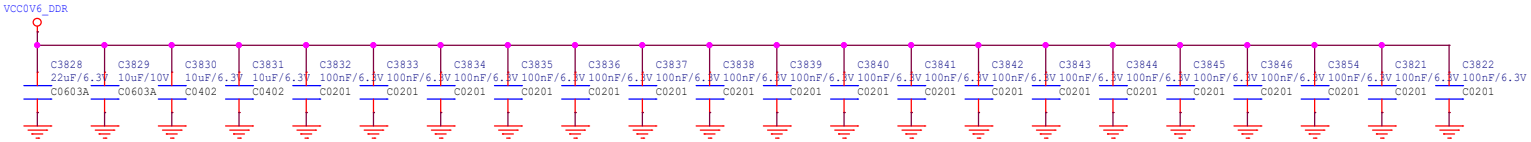

## VCC1V8_DDR




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|---|---|---|
| Size A3 | Page Name 07.DRAM-LPDDR4X_1X32bit_200P | |
| Date: Thursday, November 14, 2024 | Sheet: 7 / 21 | |

# RK3566_G(OSC/PLL/PMUIO1/2)

FLASH_VOL_SEL state decided
to VCCIO2 domain IO driven by default
Logic=L: 3.3V IO driven
Logic=H: 1.8V IO driven

FLASH_VOL_SEL R3808 1 2 2.2K R0201 VCC3V3_PMU

Adjust the load capacitor
according to the crystal spec.

U1000G

**OSC**
XOUT24M
XIN24M

R1200 22R R0201 XOUT24M AD38
R1201 1M R0201
X11 24MHz/10PPM/8pF
XIN24M AD37
C1201 15pF/50V C0201
C1202 15pF/50V C0201

Operating Voltage =1.8V Only (PMUPLL_AVDD_1V8)

1M18 TVSS
[15] RTCIC_INT_L_GPIO0_D3 1N19 GPIO0_D3_d
1M19 GPIO0_D4_d
1N20 GPIO0_D5_d
1P20 GPIO0_D6_d
DO NOT USE

**PMU PLL**
PMUPLL_AVDD_0V9
PMUPLL_AVDD_1V8
PMUPLL_AVSS

**SYS PLL**
SYSPLL_AVDD_0V9
SYSPLL_AVDD_1V8
SYSPLL_AVSS

**PMUIO1 Domain**
Operating Voltage=3.3V Only

nPOR_u AG38
REFCLK_OUT / GPIO0_A0_d
TSADC_SHUT_M0 / TSADC_SHUT_ORG / GPIO0_A1_d
PMIC_SLEEP / TSADC_SHUT_M1 / GPIO0_A2_d
GPIO0_A3_u
SDMMC0_DET / SATA_CP_DET / GPIO0_A4_u
SDMMC0_PWREN / SATA_MP_SWITCH / PCIE20_CLKREQn_M0 / GPIO0_A5_d
GPU_PWREN / SATA_CP_POD / GPIO0_A6_d
FLASH_VOL_SEL / GPIO0_A7_u
PMUIO1

**PMUIO2 Domain**
Operating Voltage=1.8V/3.3V

CLK32K_IN / CLK32K_OUT0 / GPIO0_B0_u
I2C0_SCL / GPIO0_B1_u
I2C0_SDA / GPIO0_B2_u
I2C1_SCL / MCU_JTAG_TDO / GPIO0_B3_u
I2C1_SDA / PCIE20_BUTTONRSTn / MCU_JTAG_TCK / GPIO0_B4_u
PWM1_M1 / I2C2_SCL_M0 / SPI0_CLK_M0 / PCIE20_WAKEn_M0 / GPIO0_B5_u
PWM2_M1 / I2C2_SDA_M0 / SPI0_MOSI_M0 / PCIE20_PERSTn_M0 / GPIO0_B6_u
PWM0_M0 / CPUAVS / GPIO0_B7_d
PWM1_M0 / GPUAVS / UART0_RX / GPIO0_C0_d
PWM2_M0 / NPUAVS / UART0_TX / MCU_JTAG_TDI / GPIO0_C1_d
PWM3_IR / EDP_HPDIN_M1 / MCU_JTAG_TMS / GPIO0_C2_d
PWM4 / VOP_PWM_M0 / MCU_JTAG_TRSTn / GPIO0_C3_d
PWM5 / SPI0_CS1_M0 / UART0_RTSn / GPIO0_C4_d
PWM6 / SPI0_MISO_M0 / GPIO0_C5_d
PWM7_IR / SPI0_CS0_M0 / GPIO0_C6_d
PWM0_M1 / HDMITX_CEC_M1 / UART0_CTSn / GPIO0_C7_d
UART2_RX_M0 / GPIO0_D0_u
UART2_TX_M0 / GPIO0_D1_u
PMUIO2

**PMUIO1/2/OSC Domain Logic Power**
Operating Voltage=0.9V

PMU_VDD_LOGIC_0V9

RK3566

VDDA0V9_PMU
C1204 1uF/10V C0402
C1205 100nF/6.3V C0201
1P17 PMUPLL_AVDD_0V9

VCCA1V8_PMU
C1206 1uF/10V C0402
C1207 100nF/6.3V C0201
1N18 PMUPLL_AVDD_1V8
1N17 PMUPLL_AVSS

VDDA_0V9
C1208 1uF/10V C0402
C1209 100nF/6.3V C0201
1H5 SYSPLL_AVDD_0V9

VCCA_1V8
C1211 1uF/10V C0402
C1212 100nF/6.3V C0201
1H4 SYSPLL_AVDD_1V8
1J5 SYSPLL_AVSS

Caps should be placed close to the U1000 package
Caps should be placed under the U1000 package

C1200 100nF/6.3V C0201
Caps should be placed under the U1000 package

1P18 1R20 TSADC_SHUT_M0 R1202 0R R0201 RESETn [14,20]
1P19 PMIC_SLEEP_H [14,15]
1R19 PMIC_INT_L [14]
AF37 SDMMC0_DET_L [17]
AG37 AJ38 USB_HOST_PWREN_H GPIO0_A6 [20]
AF38 FLASH_VOL_SEL

VCC3V3_PMU
1P16 C1203 100nF/6.3V C0201
PMUIO1

VCC3V3_PMU
R1203 2.2K R0201
R1204 2.2K R0201
I2C0_SCL_PMIC
I2C0_SDA_PMIC

1T20 AJ37 I2C0_SCL_PMIC [14,15]
AK38 I2C0_SDA_PMIC [14,15]
AK37 I2C1_SCL [15]
AM38 I2C1_SDA [15]
1U20 I2C2_SCL_M0 [20]
1R16 I2C2_SDA_M0 [20]
1R17 GPIO0_B7_LED_SYS [20]

VCC3V3_PMU
R3819 2.2K R0201
R3820 2.2K R0201
I2C1_SCL
I2C1_SDA

1T19 1R18 AM37 GPIO0_C2_K1 [20]
AN38 PWM4_LCD_BL [20]
AN37 GPIO0_C4_LCD_RST [20]
1U19 GPIO0_C5_TP_RST [20]
1T18 GPIO0_C6_TP_INT [20]
1V20

1V19 UART2_RX_M0_DEBUG [15]
1U18 UART2_TX_M0_DEBUG [15]

VCC3V3_PMU
R3881 2.2K R0201
R3882 2.2K R0201
I2C2_SCL_M0
I2C2_SDA_M0

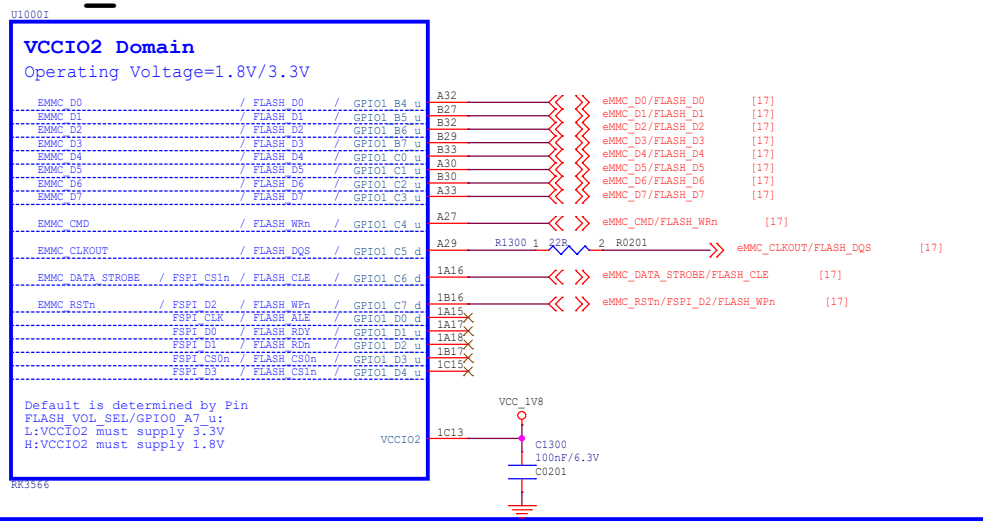
VCC3V3_PMU
1N15 C1210 100nF/6.3V C0201
1N16 C1213 100nF/6.3V C0201
Caps should be placed under the U1000 package

VDDA0V9_PMU
C1214 1uF/10V C0402
Caps should be placed close to the U1000 package

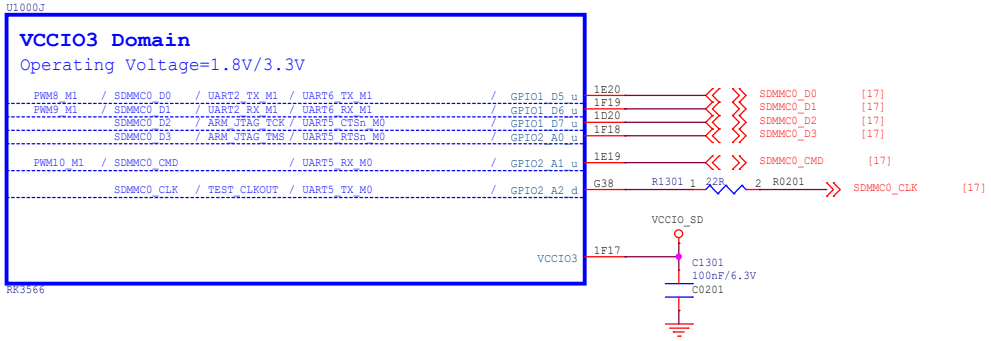
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|---|---|
| Size A3 | Page Name 08.RK3566_OSC/PLL/PMUIO |
| Date: Tuesday, October 29, 2024 | Sheet: 8 / 21 |

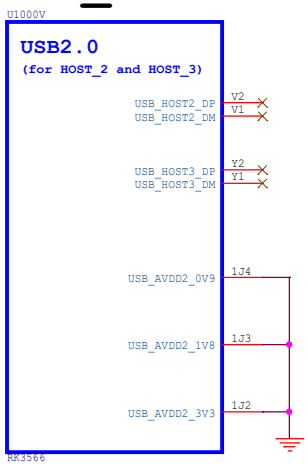
# RK3566_I(VCCIO2 Domain)



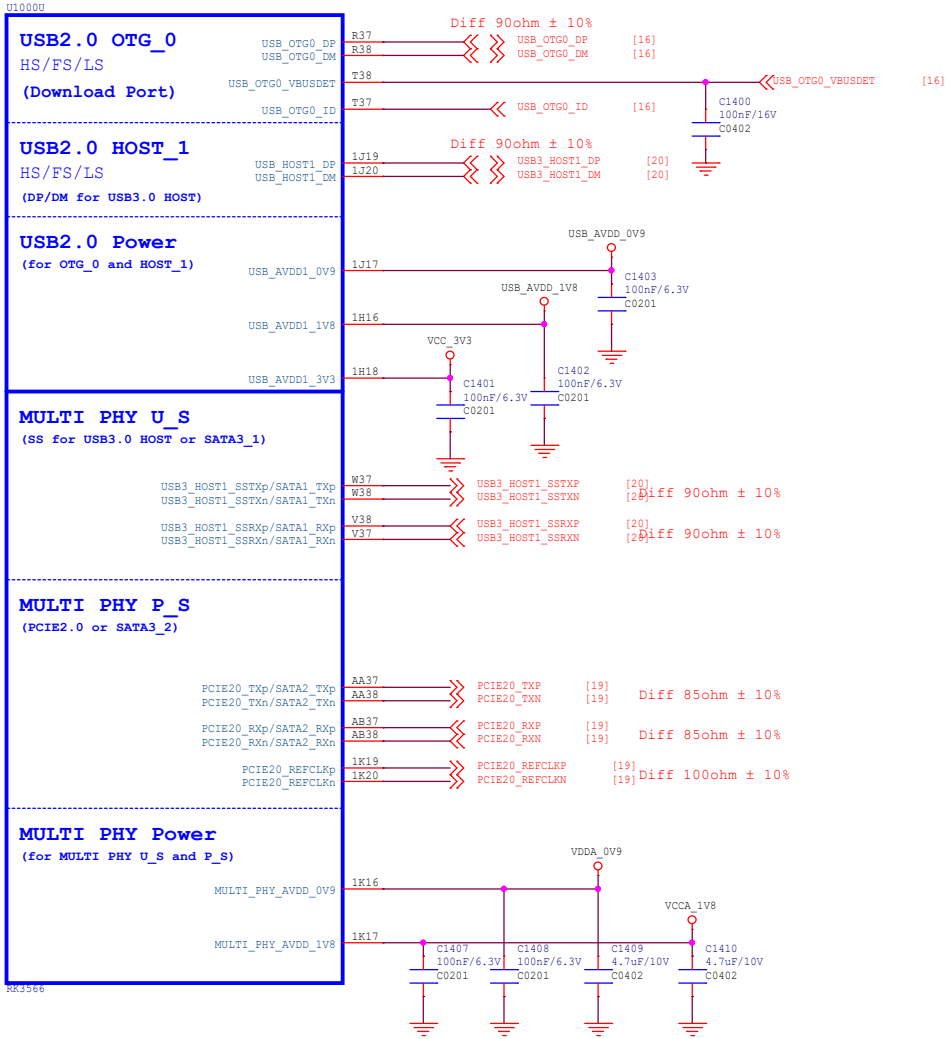

# RK3566_J(VCCIO3 Domain)




# RK3566_V(USB2.0 HOST)



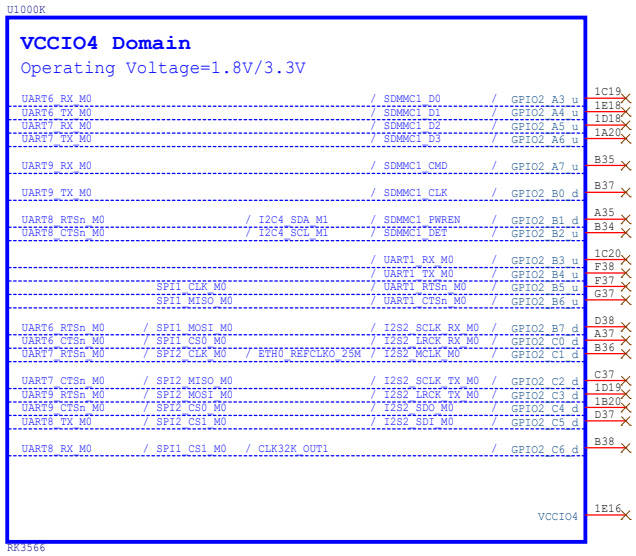

# RK3566_U(USB3.0/PCIe2.0x1/SATA)



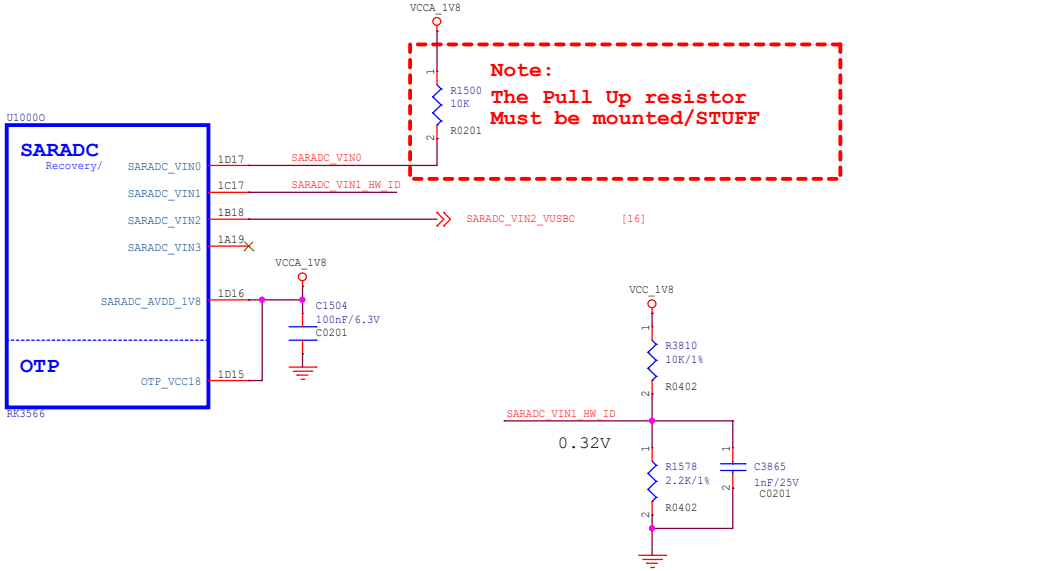

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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 09.RK3566_Flash/SD/USB/PCIe | |
| Date: Tuesday, October 29, 2024 | Sheet: 9 / 21 | |

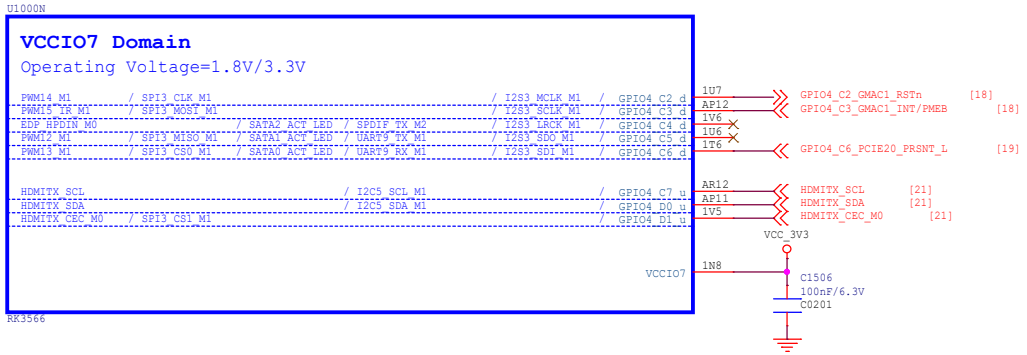
# RK3566_K(VCCIO4 Domain)



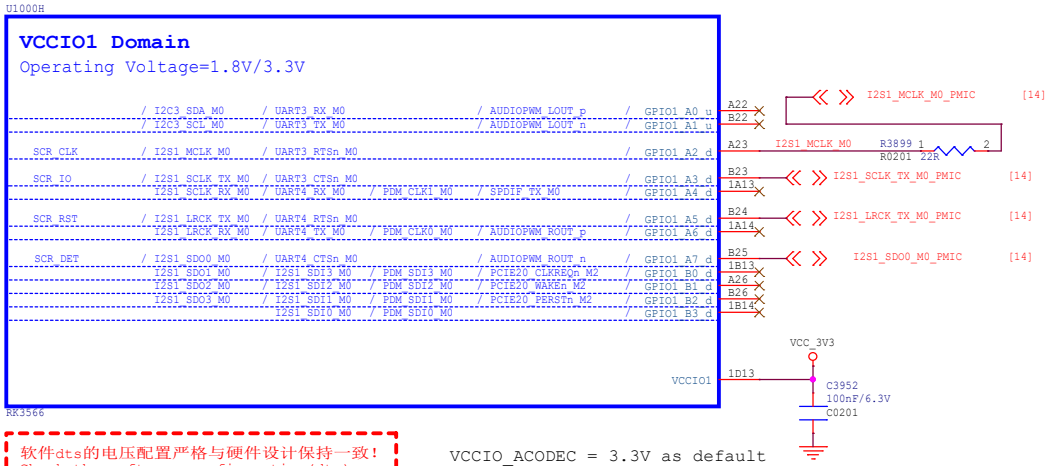

# RK3566_O(SARADC/OTP)




# RK3566_N(VCCIO7 Domain)




# RK3566_H(VCCIO1 Domain)



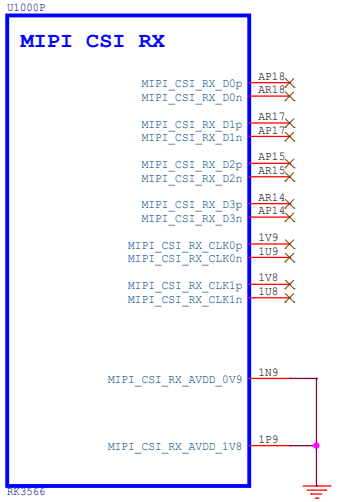

软件dts的电压配置严格与硬件设计保持一致!
Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design

VCCIO_ACODEC = 3.3V as default

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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 10.RK3566_SARADC/GPIO | |
| Date: Friday, November 15, 2024 | Sheet: 10/ 21 | |

# RK3566_P(MIPI_CSI_RX)



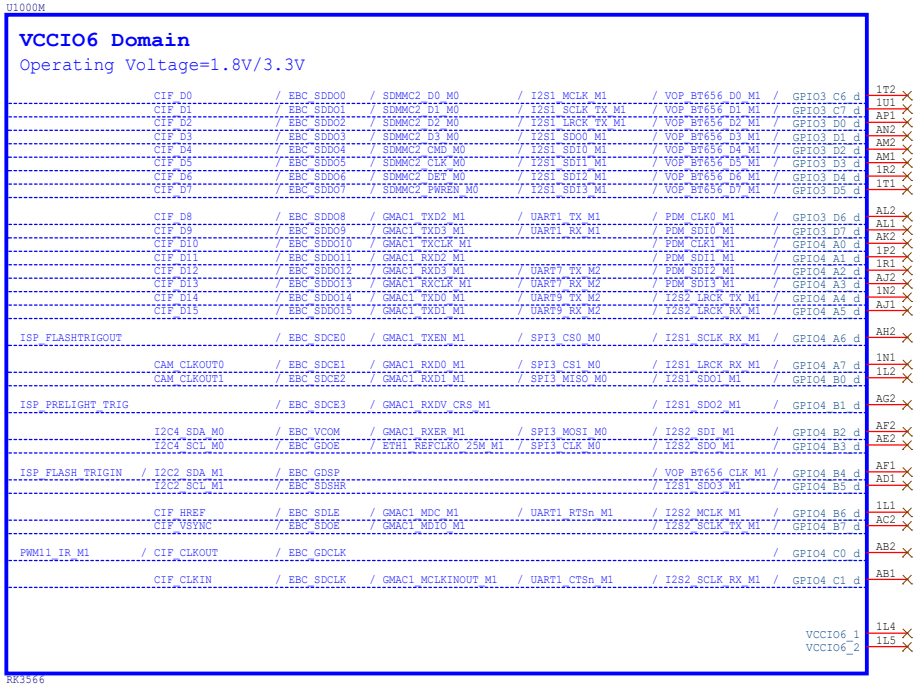

| Usage of MIPI CSI Dx&CLKs | | |
|---|---|---|
| Option1 | Sensor1 x4Lane | MIPI_CSI_RX_D0-3<br>MIPI_CSI_RX_CLK0 |
| Option2 | Sensor1 x2Lane<br>+<br>Sensor2 x2Lane | MIPI_CSI_RX_D0-1<br>MIPI_CSI_RX_CLK0<br>MIPI_CSI_RX_D2-3<br>MIPI_CSI_RX_CLK1 |

| Usage of CIF Interface | | | | |
|---|---|---|---|---|
| Mode | 16bit | 12bit | 10bit | 8bit |
| CIF_D0 | D0 | -- | -- | -- |
| CIF_D1 | D1 | -- | -- | -- |
| CIF_D2 | D2 | -- | -- | -- |
| CIF_D3 | D3 | -- | -- | -- |
| CIF_D4 | D4 | D0 | -- | -- |
| CIF_D5 | D5 | D1 | -- | -- |
| CIF_D6 | D6 | D2 | D0 | -- |
| CIF_D7 | D7 | D3 | D1 | -- |
| CIF_D8 | D8 | D4 | D2 | D0 |
| CIF_D9 | D9 | D5 | D3 | D1 |
| CIF_D10 | D10 | D6 | D4 | D2 |
| CIF_D11 | D11 | D7 | D5 | D3 |
| CIF_D12 | D12 | D8 | D6 | D4 |
| CIF_D13 | D13 | D9 | D7 | D5 |
| CIF_D14 | D14 | D10 | D8 | D6 |
| CIF_D15 | D15 | D11 | D9 | D7 |

Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 8/10/12/16bit input, single/dual-edge sampling
Support 2/4 mixed BT656/BT1120 YCbCr 422 8bit input

BT1120 16bit Mode:
Default: D0-D7 <--> Y0-Y7 , D8-D15 <--> C0-C7
Swap ON: D0-D7 <--> C0-C7 , D8-D15 <--> Y0-Y7

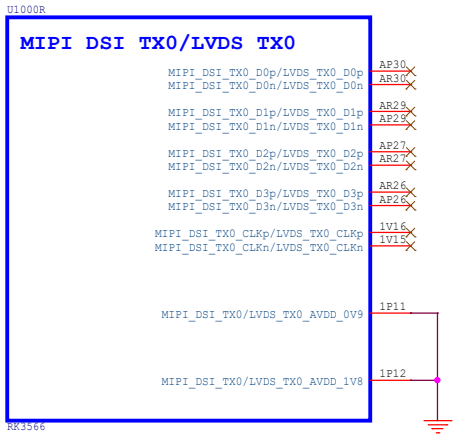
# RK3566_M(VCCIO6 Domain)



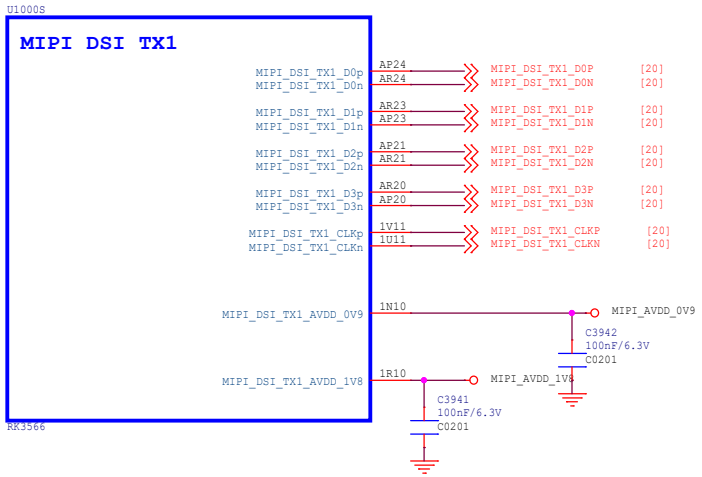

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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 11.RK3566_VI_Interface | |
| Date: Tuesday, October 29, 2024 | Sheet: 11/ 21 | |

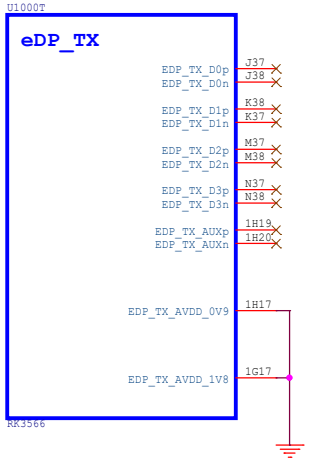
# RK3566_R(MIPI_DSI_TX0/LVDS_TX0)



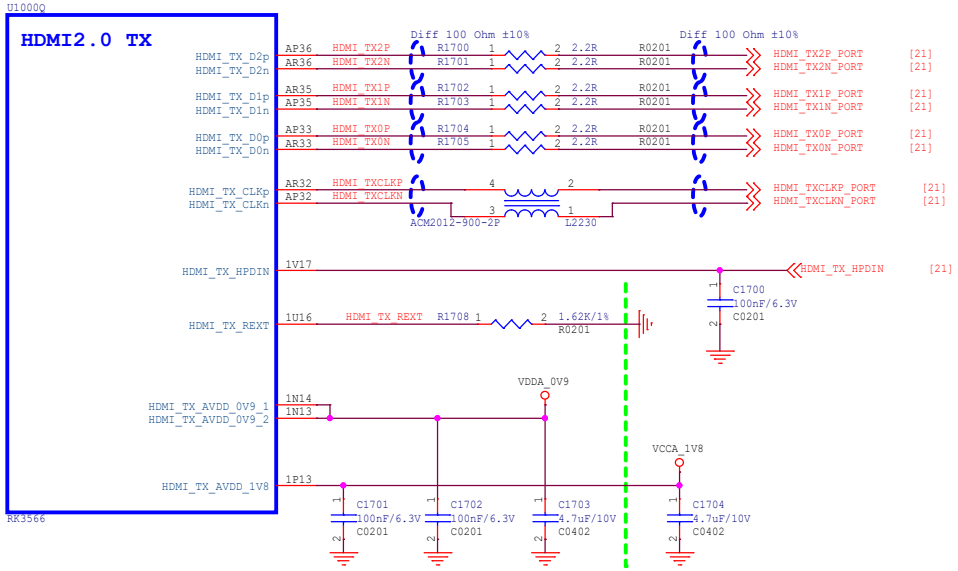

# RK3566_S(MIPI_DSI_TX1)




# RK3566_T(eDP TX)



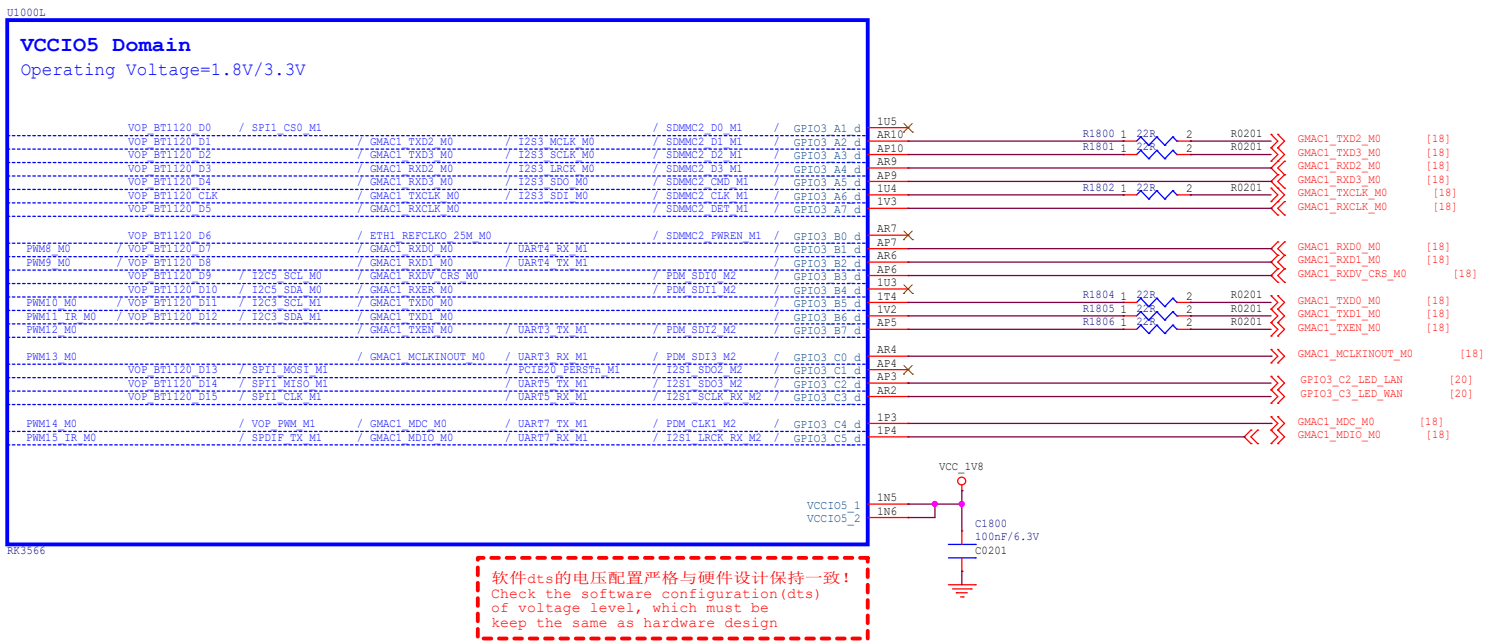

# RK3566_Q(HDMI2.0 TX)




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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 12.RK3566_VO_Interface_1 | |
| Date: Monday, November 11, 2024 | Sheet: 12/ 21 | |

# RK3566_L(VCCIO5 Domain)



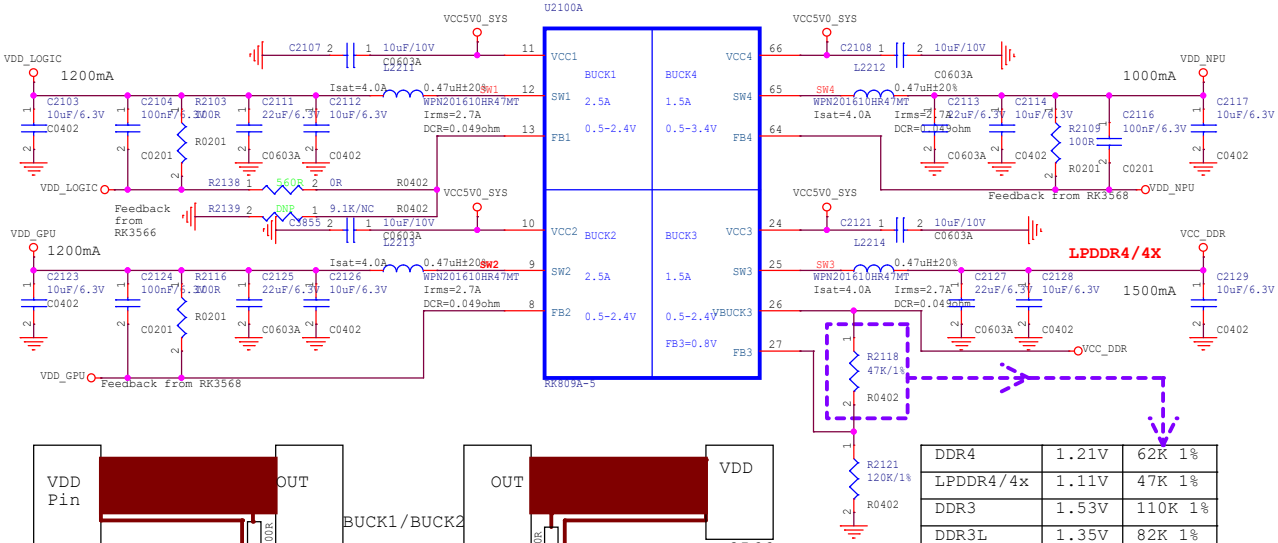

软件dts的电压配置严格与硬件设计保持一致！
Check the software configuration(dts) of voltage level, which must be keep the same as hardware design

| GMAC | Direction | GEPHY | GMAC | Direction | FEPHY |
|---|---|---|---|---|---|
| GMACx_TXD0 | ------> | PHYx_TXD0 | GMACx_TXD0 | ------> | PHYx_TXD0 |
| GMACx_TXD1 | ------> | PHYx_TXD1 | GMACx_TXD1 | ------> | PHYx_TXD1 |
| GMACx_TXD2 | ------> | PHYx_TXD2 | | | |
| GMACx_TXD3 | ------> | PHYx_TXD3 | | | |
| GMACx_TXEN | ------> | PHYx_TXEN | GMACx_TXEN | ------> | PHYx_TXEN |
| GMACx_TXCLK | ------> | PHYx_TXCLK | | | |
| GMACx_RXD0 | <------ | PHYx_RXD0 | GMACx_RXD0 | <------ | PHYx_RXD0 |
| GMACx_RXD1 | <------ | PHYx_RXD1 | GMACx_RXD1 | <------ | PHYx_RXD1 |
| GMACx_RXD2 | <------ | PHYx_RXD2 | | | |
| GMACx_RXD3 | <------ | PHYx_RXD3 | | | |
| GMACx_RXDV | <------ | PHYx_RXDV | GMACx_RXDV | <------ | PHYx_CRS_DV |
| GMACx_RXCLK | <------ | PHYx_RXCLK | | | |
| GMACx_RXER | | | GMACx_RXER | <------ | PHYx_RXER |
| GMACx_MDC | ------> | PHYx_MDC | GMACx_MDC | ------> | PHYx_MDC |
| GMACx_MDIO | <-----> | PHYx_MDIO | GMACx_MDIO | <-----> | PHYx_MDIO |
| ETHx_REFCLKO_25M | ------> | PHYx_XTALIN | | | |
| GMACx_MCLKINOUT | <------ | PHYx_CLKOUT125 (Option) | GMACx_MCLKINOUT | ------> | PHYx_XTALIN/REFCLK |
| GPIO | ------> | PHYx_RSTn | GPIO | ------> | PHYx_RSTn |
| GPIO | <------ | PHYx_INT/PMEB | GPIO | <------ | PHYx_INT/PMEB |

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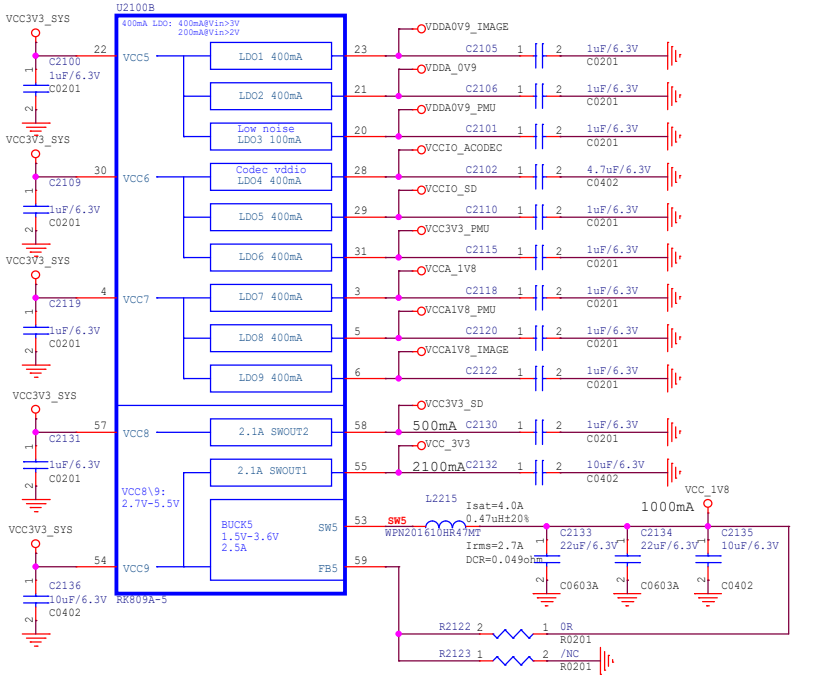
| NanoPi R3S LTS | Rev 2411 |
|---|---|
| Size A3 | Page Name 13.RK3566_VO_Interface_2 |
| Date: Tuesday, October 29, 2024 | Sheet: 13/ 21 |

# PMIC RK809 DCDC

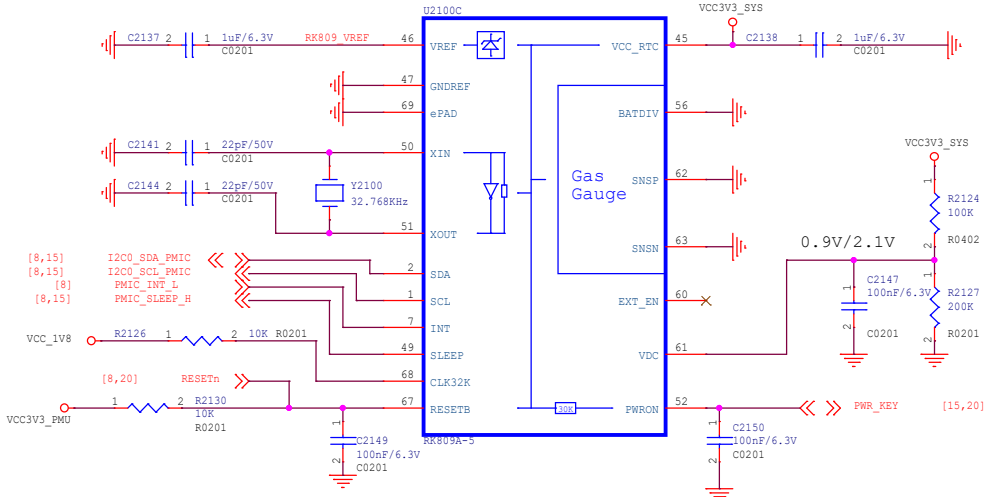



| | | |
|---|---|---|
| DDR4 | 1.21V | 62K 1% |
| LPDDR4/4x | 1.11V | 47K 1% |
| DDR3 | 1.53V | 110K 1% |
| DDR3L | 1.35V | 82K 1% |
| LPDDR3 | 1.25V | 68K 1% |

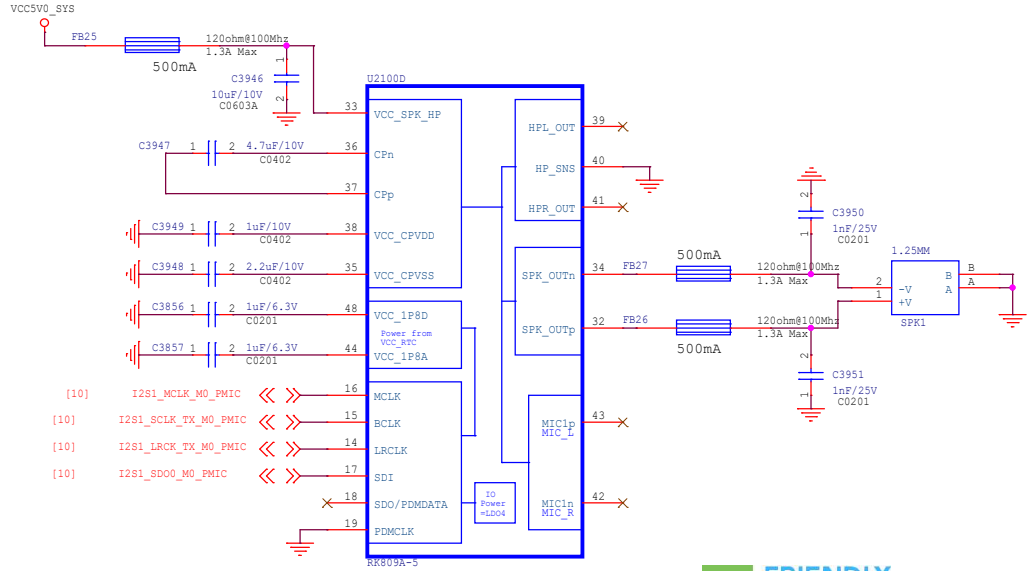
# PMIC RK809 LDO




# PMIC RK809 Managerment




# PMIC RK809 CODEC

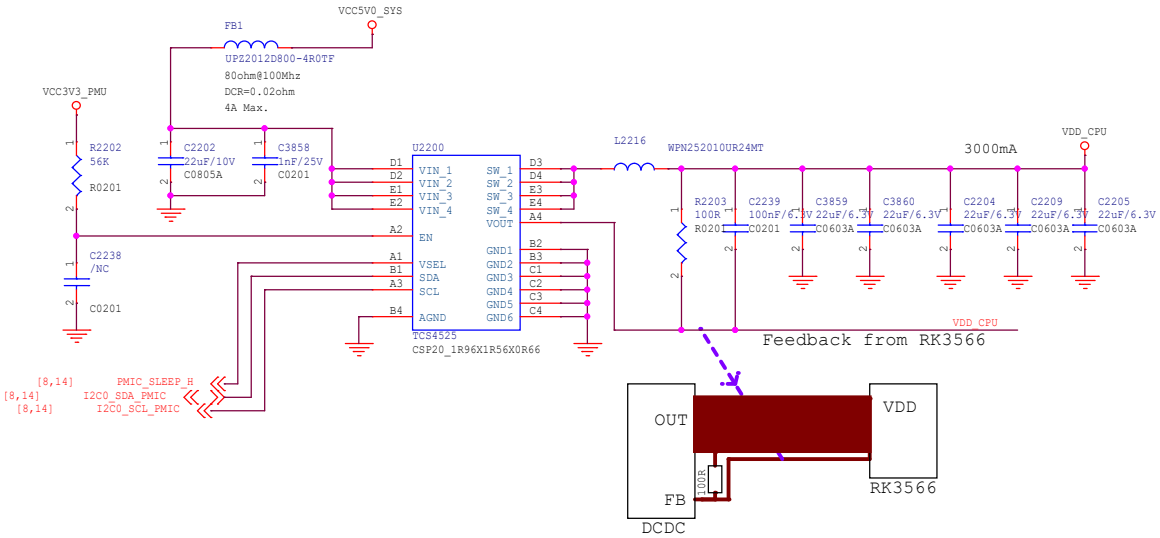



Note:
If RK809-5 codec is not used,
then Pin 14,15,16,17,19,40 Tie VSS
Pin 18,36,37,38,35,39,41,34,32,43,42
Leave floating

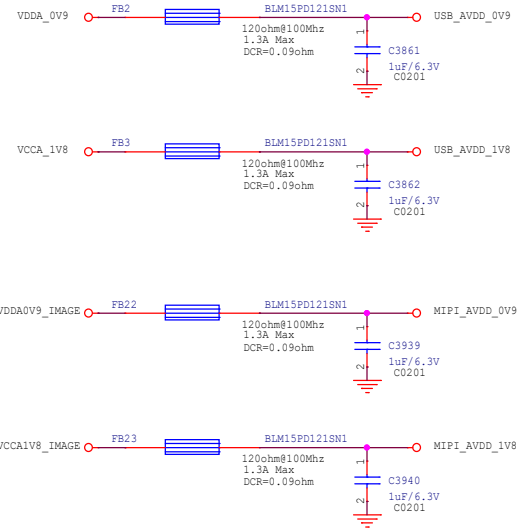
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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 14.Power_PMIC | |
| Date: Thursday, November 14, 2024 | Sheet: 14/ 21 | |

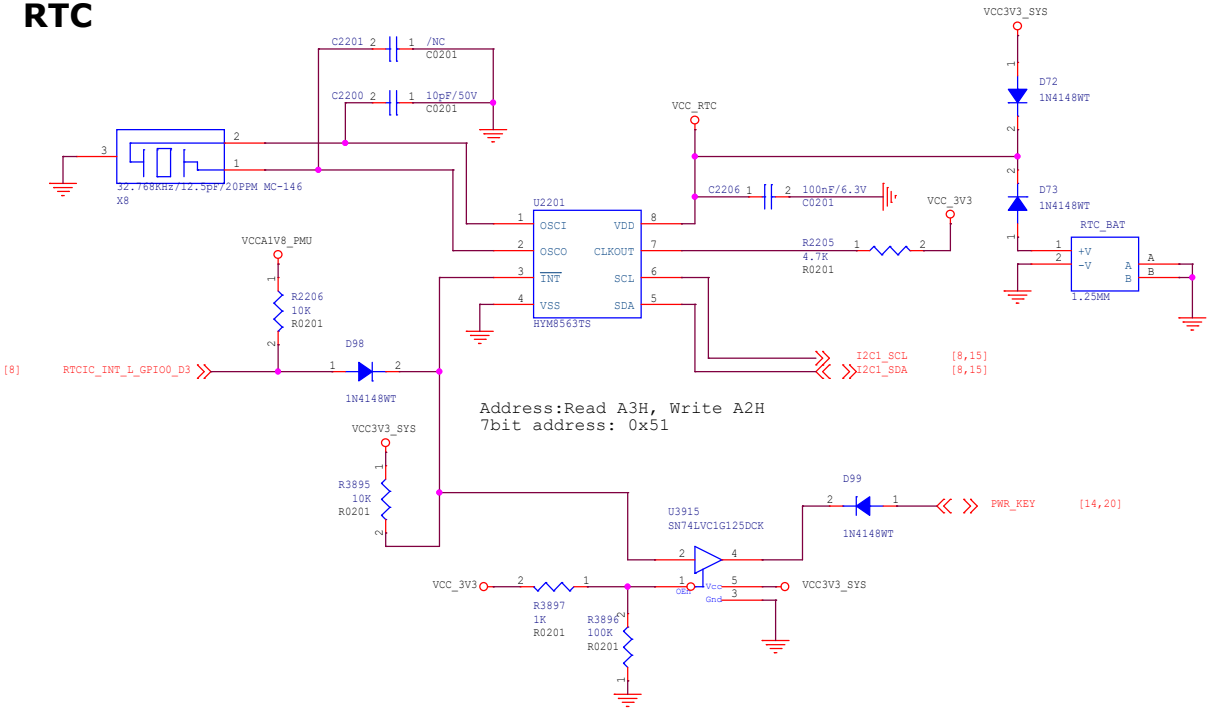
## VDD_CPU



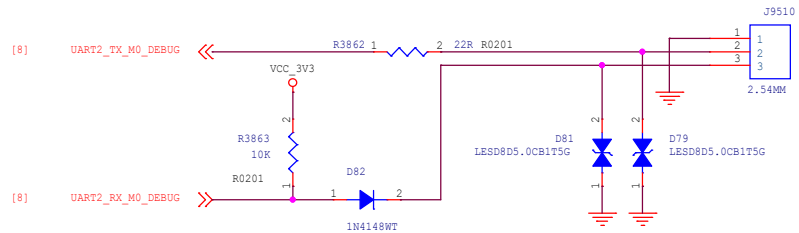

## VDD_USB

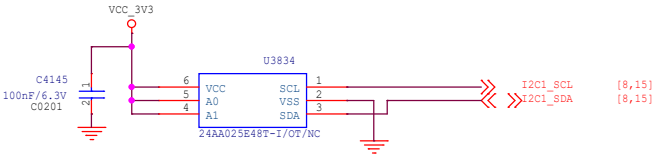



## RTC




## Debug UART



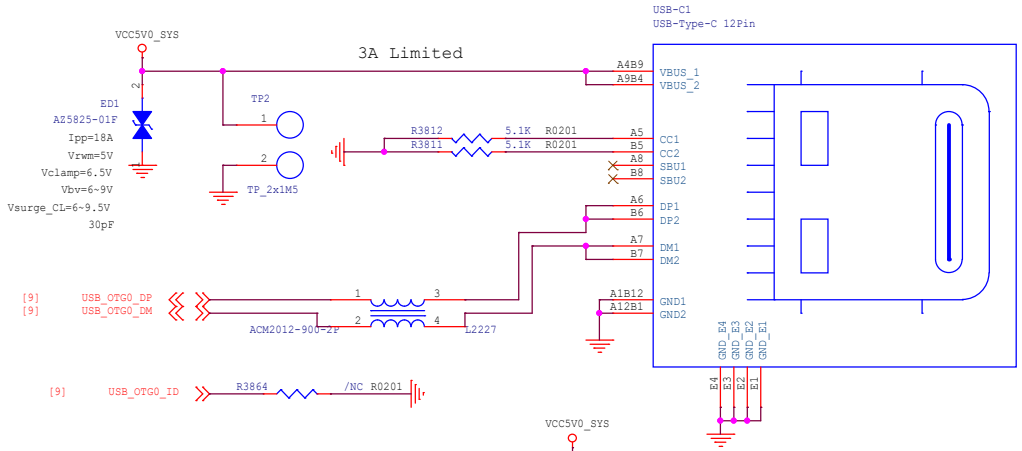



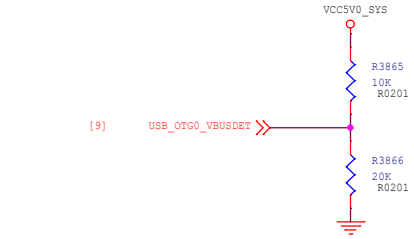

7 bit address: 0x53

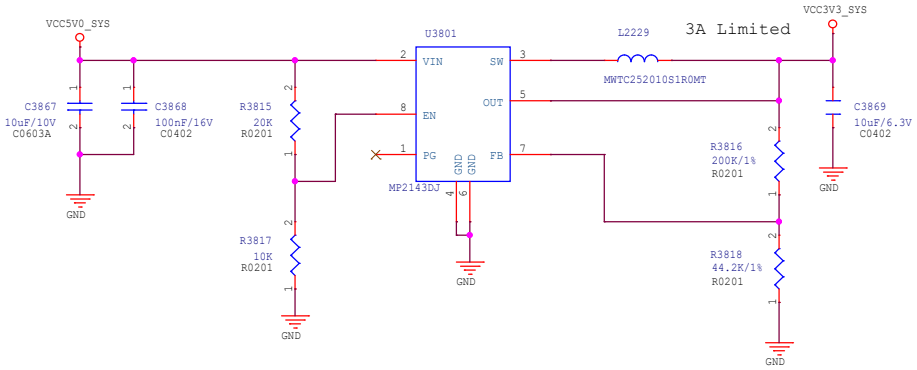
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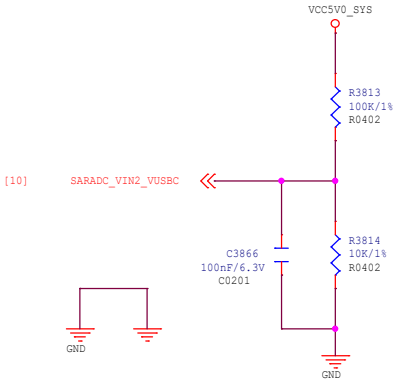
| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 15.Power_other/RTC/Debug | |
| Date: Tuesday, October 29, 2024 | Sheet: 15/ 21 | |

# Power IN



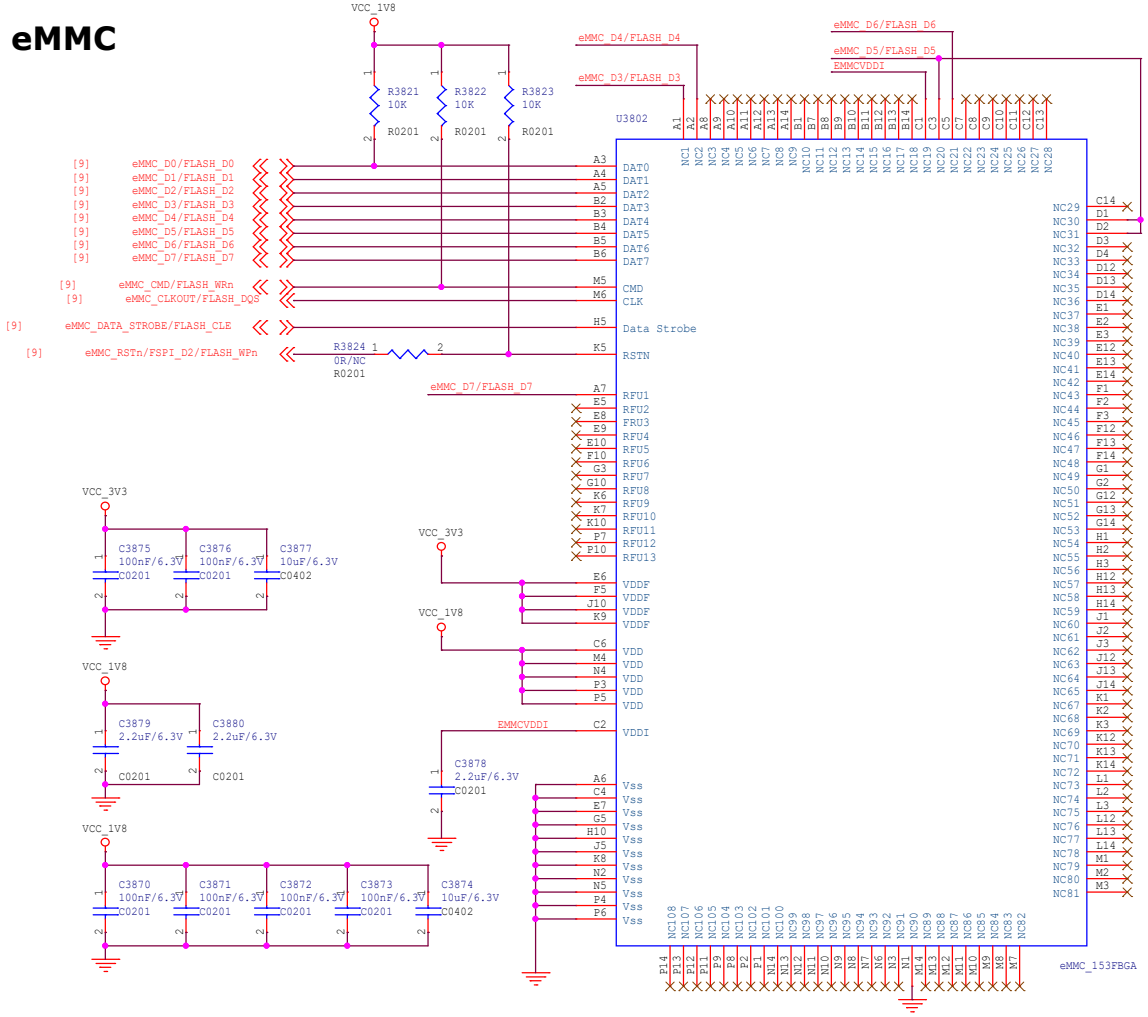



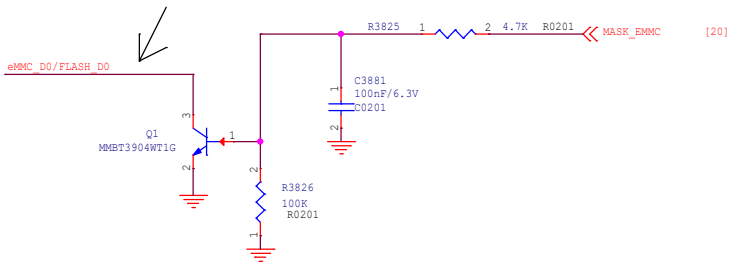

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| NanoPi R3S LTS | Rev 2411 |
|---|---|
| Size A3 | Page Name 16.Power IN |
| Date: Tuesday, October 29, 2024 | Sheet: 16/ 21 |

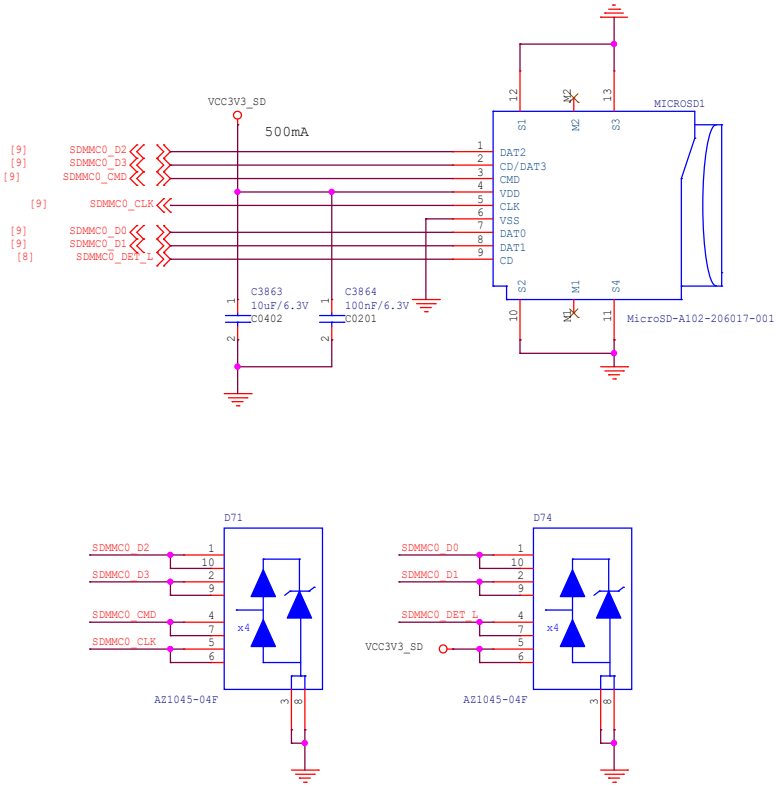
# eMMC




Stub as short as possible



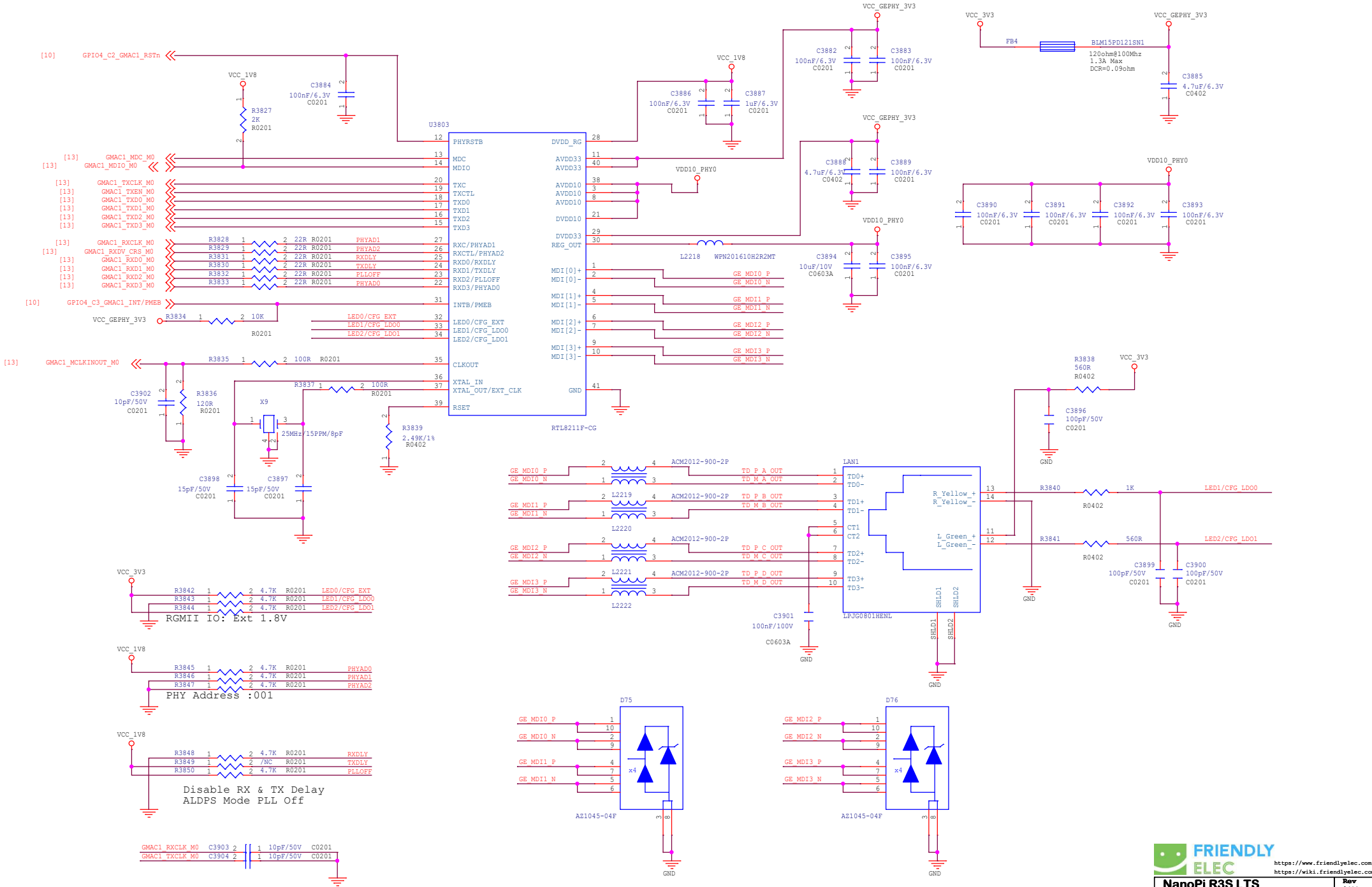

# microSD




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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 17.eMMC/SD | |
| Date: Tuesday, October 29, 2024 | Sheet: 17/ 21 | |

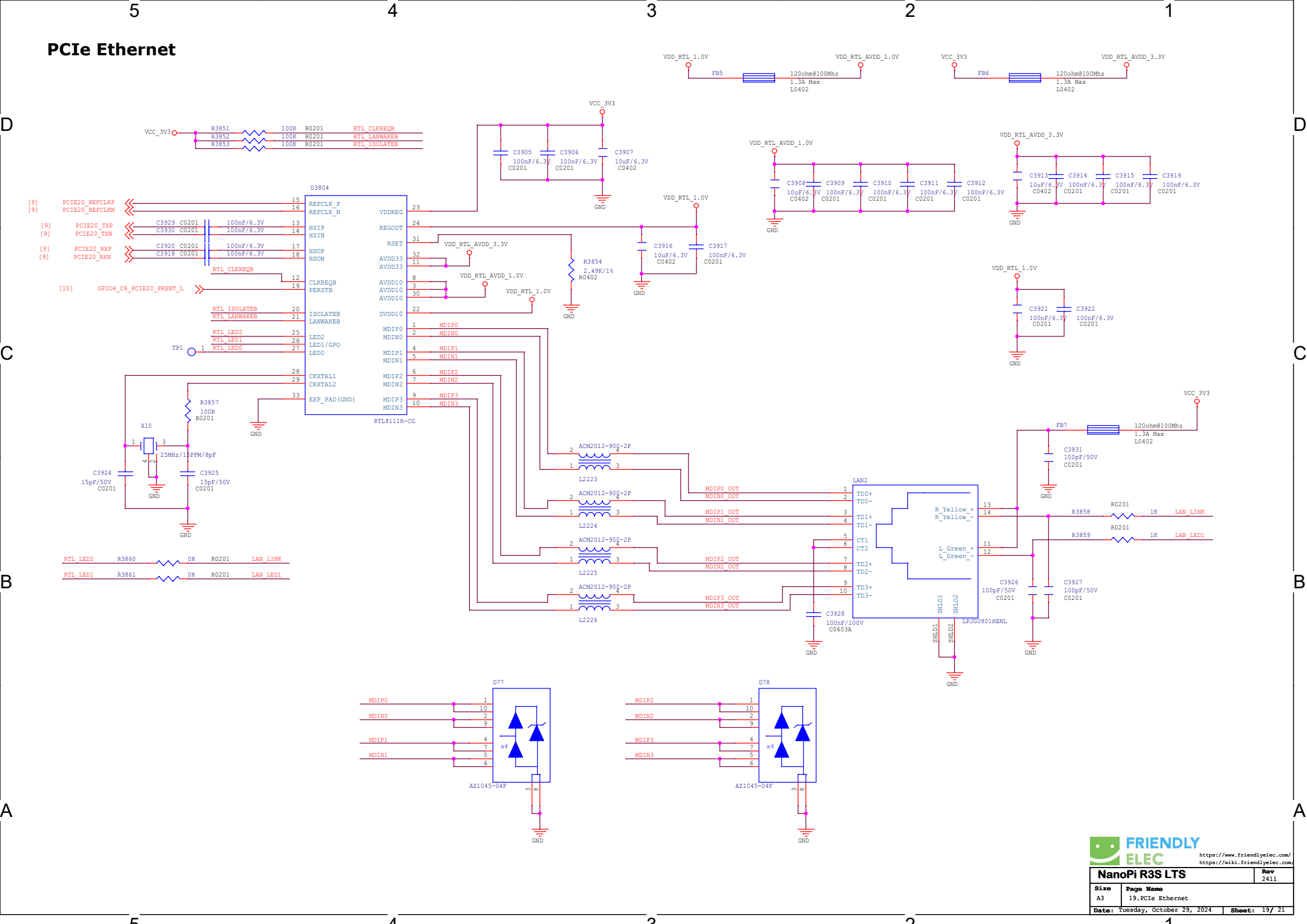
# GMAC Ethernet




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| NanoPi R3S LTS | Rev 2411 |
|---|---|
| Size A3 | Page Name 18.GMAC Ethernet |
| Date: Tuesday, October 29, 2024 | Sheet: 18/ 21 |

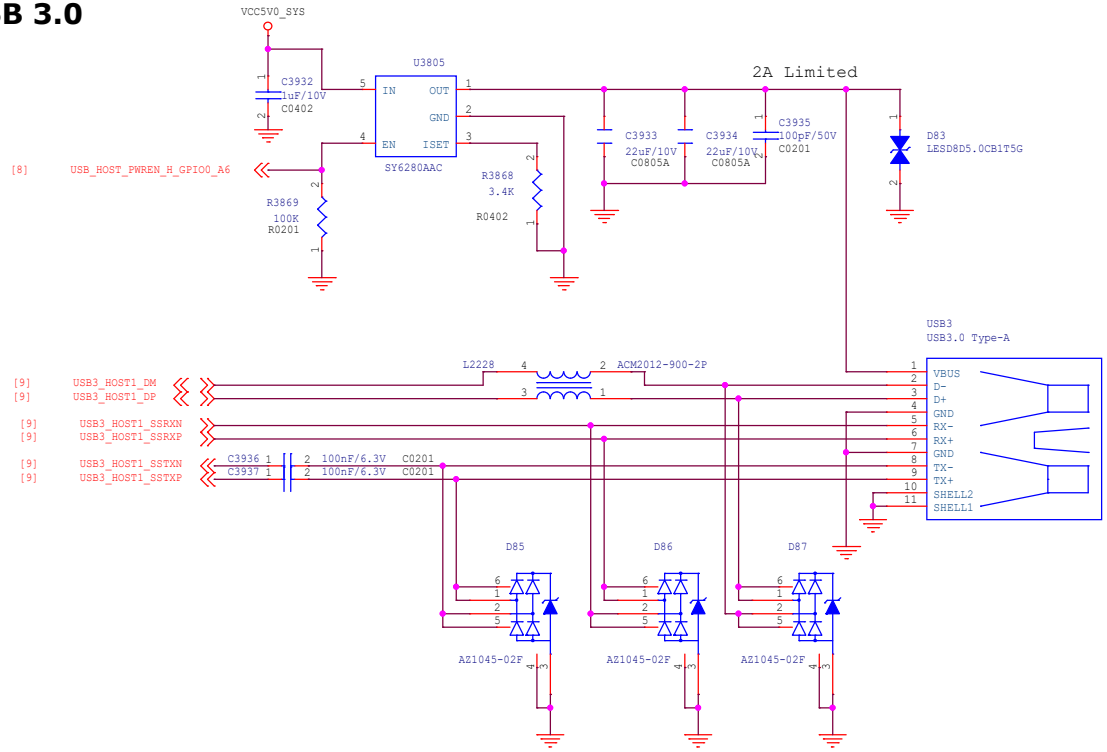
# PCIe Ethernet



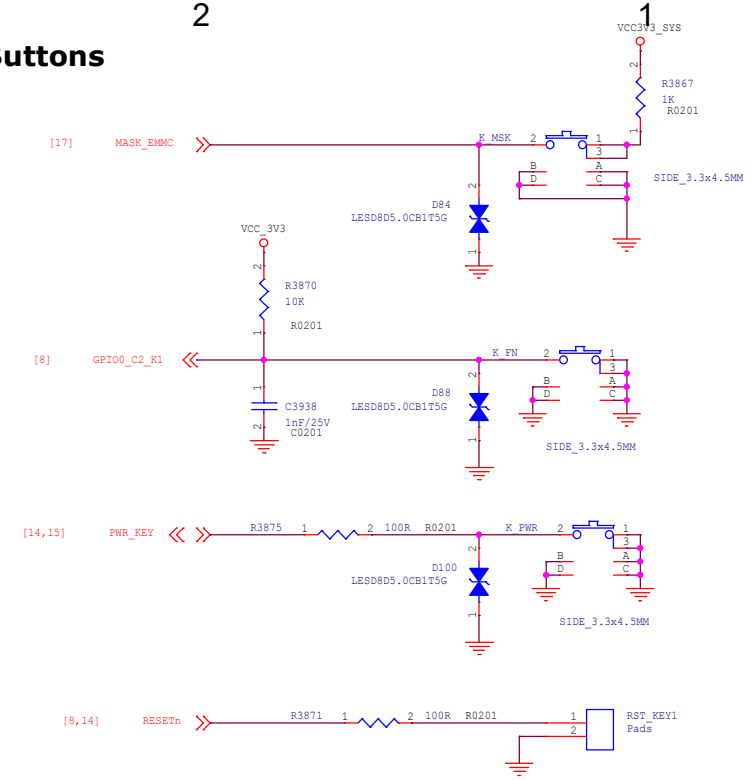

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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 19.PCIe Ethernet | |
| Date: Tuesday, October 29, 2024 | | Sheet: 19/ 21 |

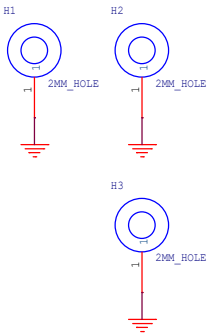
## USB 3.0



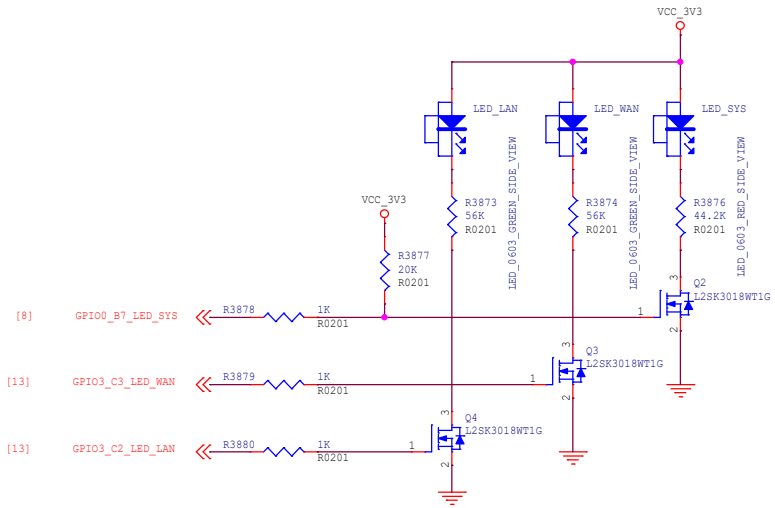

## Buttons



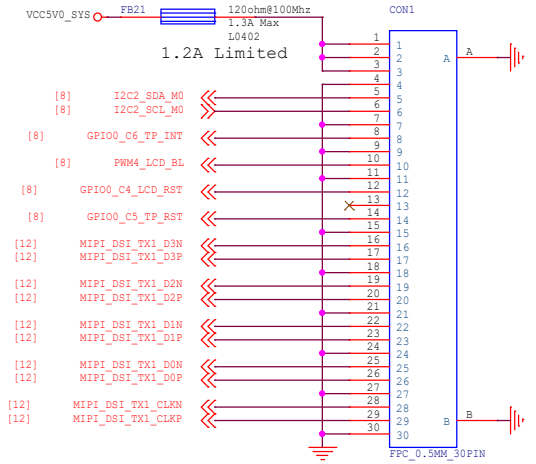

## Screw Holes




## LEDs



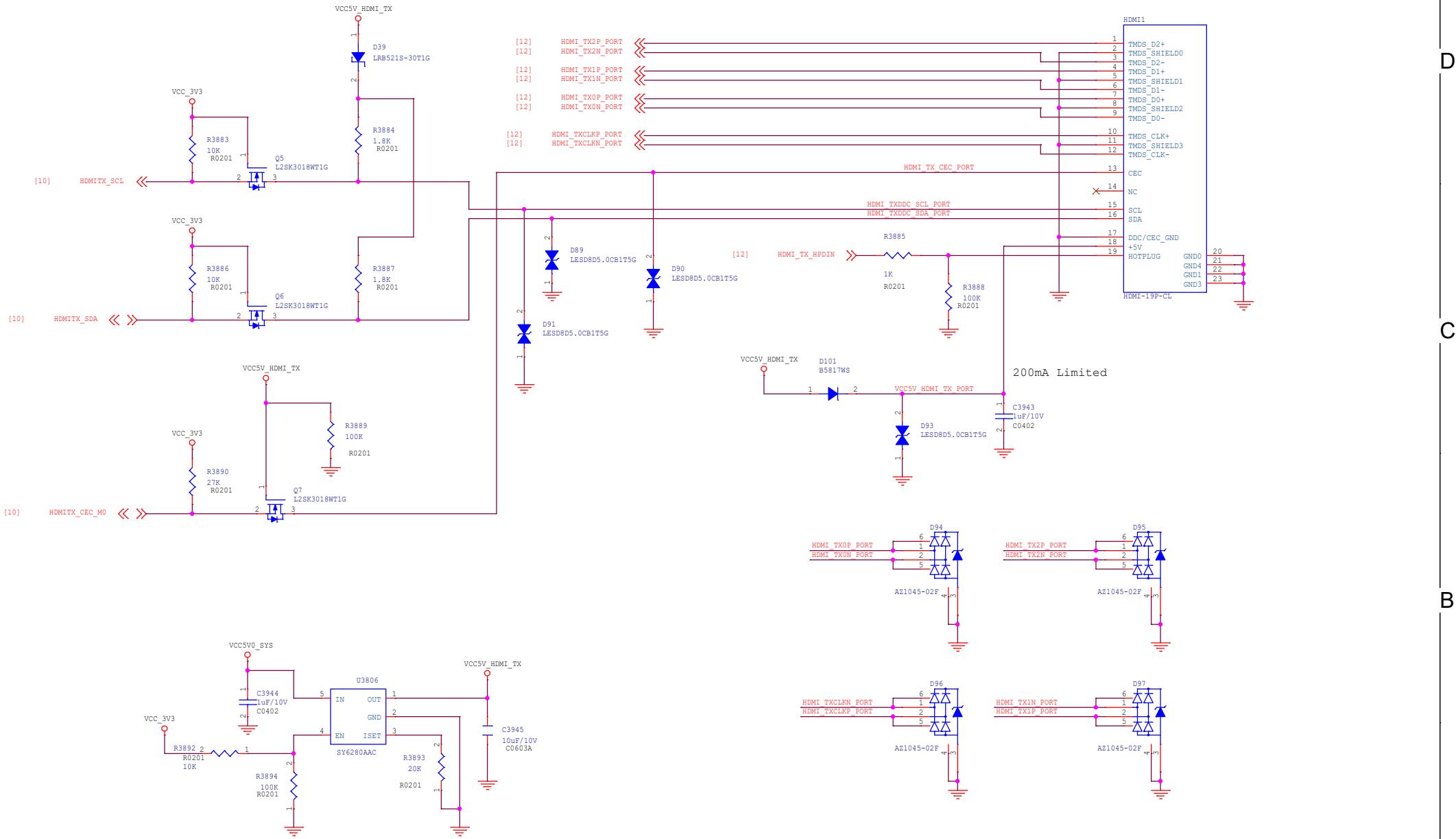

## MIPI LCD




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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 20.USB3.0/Buttons/LEDs/LCD | |
| Date: Monday, November 11, 2024 | Sheet: 20/ 21 | |

# HDMI 2.0 TX




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| NanoPi R3S LTS | | Rev 2411 |
|---|---|---|
| Size A3 | Page Name 21.HDMI | |
| Date: Thursday, November 07, 2024 | Sheet: 21/ 21 | |